

Year Up United

Data Analyst Training Academy:

Introduction to Power BI

Week 9 Student Guide

This courseware is to be distributed to accompany instructor led training.

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Week 9 Learning Objectives

- This week you will learn:
 - ◇ Power BI's role in the data analysis workflow and how it relates to the tools used earlier in the course (Excel, SQL, Python)
 - ◇ Power BI Desktop interface: home screen, the four views (Report, Table, Model, DAX Query), and the context-sensitive ribbon
 - ◇ The data modeling pipeline: discover, ingest, profile, transform, clean, and augment
 - ◇ Data modeling concepts: the semantic model, data tables (fact) vs. lookup tables (dimension), star and snowflake schema rules
 - ◇ Model relationships: cardinality and filter direction, and the rules for connecting data and lookup tables
 - ◇ Column configuration: data types, numeric and date formatting, data categories, default summarization, sort order, and navigational hierarchies
 - ◇ Power Query Editor: purpose, GUI tour, supported data sources, and the step-based transformation pipeline
 - ◇ Simple Power Query transformations: renaming and removing columns, changing data types, and adding calculated fields
 - ◇ Introduction to DAX concepts: implicit vs. explicit calculations, using the formula bar, Table[Column] syntax, and creating measures and columns
 - ◇ Visualization fundamentals: selecting an appropriate chart type, matching visual to data story, basic formatting considerations, and the primary visualization types available in Power BI

Week 9 Learning Outcomes

- By the end of this week, you will be able to:
 - ◇ Describe the different components of the Power BI ecosystem and explain the role of each
 - ◇ Navigate Power BI Desktop, locate core tools and functions, and appropriately use the Report View, Table View, Model View, and DAX query view
 - ◇ Import a data source using Power Query and apply basic transformations: rename, remove, retype, and add a calculated field
 - ◇ Configure a semantic model with correctly named data and lookup tables, appropriate data types, and at least one defined relationship
 - ◇ Write a basic DAX measure using a common aggregation function (e.g. SUM, COUNT, or AVERAGE)
 - ◇ Select appropriate chart types for use in a Power BI report

Module 1

Introduction to Power BI

Section 1.A

Getting Started with Power BI

1.A.1 What is Power BI?

- Power BI is a business intelligence solution from Microsoft that enables end-to-end management of reporting, including:
 - ◇ Ingestion of data from a wide variety of sources
 - ◇ Data cleaning
 - ◇ Transformation and reshaping of data
 - ◇ Data modeling
 - ◇ Data enhancement
 - ◇ Analysis and visualization

1.A.2 Power BI Ecosystem

- Power BI consists of several elements that all work together:
 - ◇ Windows desktop application called **Power BI Desktop**
 - ◇ Online software as a service (SaaS) version called the **Power BI service**
 - ◇ Power Query Editor (within Power BI Desktop / Power BI service)
 - ◇ Power BI Mobile apps for Windows, iOS, and Android devices
- Power BI is also part of a wider Microsoft ecosystem that consists of multiple programs and tools, both on-premises and cloud-based

1.A.3 Power BI Desktop

- Free and powerful desktop application for working with Power BI semantic models and authoring reports
 - ◇ For Microsoft's own introduction, see Power BI on Microsoft Learn: <https://learn.microsoft.com/en-us/power-bi/fundamentals/power-bi-overview>
- Hosts the data in-memory using a powerful database engine called xVelocity
- Saves analysis projects as .pbix files
 - ◇ Data within the .pbix files needs to be “refreshed” or it will go stale
 - ◇ Can be problematic when sharing files, as users often have different credentials to remote data sources
 - ◇ File versioning is *not* saved with .pbix files (unlike other Microsoft products like Word or Excel)

1.A.4 Power Query and M Language

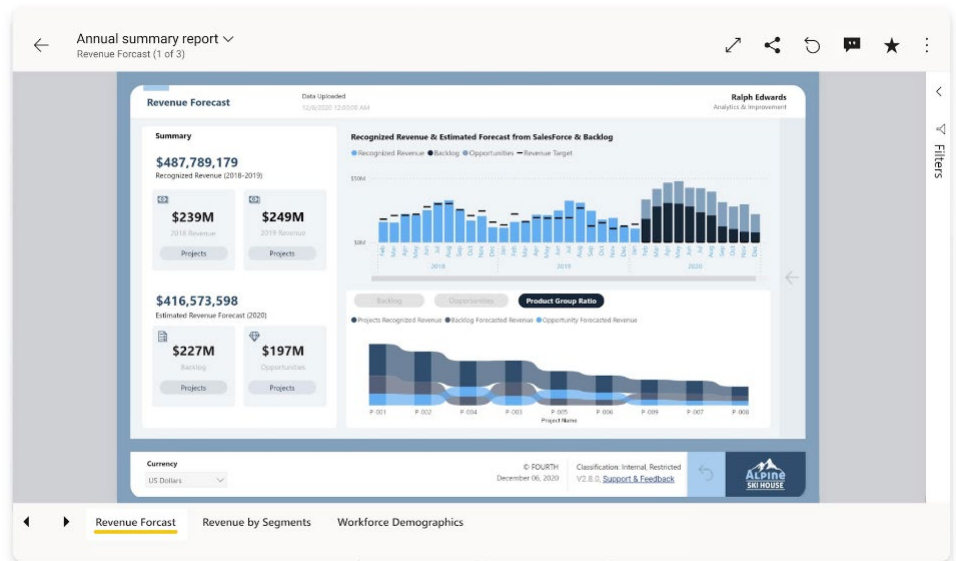
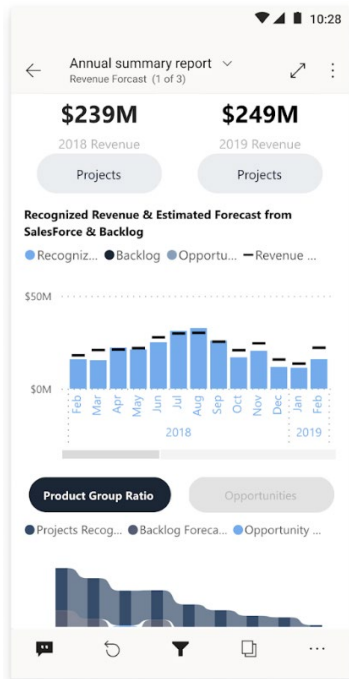
- Power Query Editor is embedded into Power BI, and provides tooling to bring data into the Power BI project to be modeled
- Data cleaning and transformation is performed in Power Query using M language

The screenshot illustrates the Power BI Desktop interface. In the top ribbon, the 'Home' tab is active, and the 'Transform data' button is highlighted with a blue box. A blue arrow points from this button to the Power Query Editor window, which is titled 'Table.AddColumn("Changed Type1", "CourseID", each "Analytics Academy | Bess Ryan | 2/6/2023")'. The editor shows a table with 21 rows and 5 columns: Email, Date, Attended Value, Attendance Type, and CourseID. The 'Applied Steps' pane on the right lists the transformations: Source, Removed Top Rows, Removed Columns, Promoted Headers, Changed Type, Filtered Rows, Unpivoted Columns, Renamed Columns1, Added Conditional Column, Removed Columns1, Changed Type1, and Added Custom. The bottom status bar indicates '5 COLUMNS, 300 ROWS' and 'Column profiling based on top 1000 rows'.

1	2	3	4	5
Email	Date	1.2 Attended Value	Attendance Type	CourseID
MichaelScott@dundermifflin.com	2/6/2023	1	Attended	Analytics Academy E
MichaelScott@dundermifflin.com	2/7/2023	1	Attended	Analytics Academy E
MichaelScott@dundermifflin.com	2/8/2023	1	Attended	Analytics Academy E
MichaelScott@dundermifflin.com	2/9/2023	1	Attended	Analytics Academy E
MichaelScott@dundermifflin.com	2/10/2023	1	Attended	Analytics Academy E
MichaelScott@dundermifflin.com	2/13/2023	1	Attended	Analytics Academy E
MichaelScott@dundermifflin.com	2/14/2023	1	Attended	Analytics Academy E
MichaelScott@dundermifflin.com	2/15/2023	1	Attended	Analytics Academy E
MichaelScott@dundermifflin.com	2/16/2023	1	Attended	Analytics Academy E
MichaelScott@dundermifflin.com	2/17/2023	1	Attended	Analytics Academy E
MichaelScott@dundermifflin.com	2/20/2023	1	Attended	Analytics Academy E
MichaelScott@dundermifflin.com	2/21/2023	1	Attended	Analytics Academy E
MichaelScott@dundermifflin.com	2/22/2023	1	Attended	Analytics Academy E
MichaelScott@dundermifflin.com	2/23/2023	1	Attended	Analytics Academy E
MichaelScott@dundermifflin.com	2/24/2023	1	Attended	Analytics Academy E
DwightSchrute@dundermifflin.com	2/6/2023	0.5	Partially Attended	Analytics Academy E
DwightSchrute@dundermifflin.com	2/7/2023	1	Attended	Analytics Academy E
DwightSchrute@dundermifflin.com	2/8/2023	1	Attended	Analytics Academy E
DwightSchrute@dundermifflin.com	2/9/2023	1	Attended	Analytics Academy E
DwightSchrute@dundermifflin.com	2/10/2023	1	Attended	Analytics Academy E

1.A.5 Power BI Mobile

- View reports hosted with the Power BI Service in a mobile friendly way



- Connects to Power BI Service, or Power BI Report Server

1.A.6 xVelocity Analytical Engine

- xVelocity is the in-memory columnar database engine that underpins Power BI
 - ◇ Stores and hosts the data model
 - ◇ DAX was written specifically as an expression language for xVelocity
- Known as Power Pivot from the perspective of Excel
 - ◇ Also referred to by its codename, VertiPaq
- Used by multiple BI related Microsoft Products
 - ◇ Power BI Desktop
 - * Hosts your model on the desktop
 - ◇ Power BI Service
 - * Hosts your model in the cloud
 - ◇ SQL Server Analysis Services Tabular Engine
 - ◇ Power Pivot for Excel

1.A.7 Columnar Databases

- Columnar databases are ideal for analytical reporting
 - ◇ Computers read and write from disks in large chunks, often 64k bytes at a time
 - ◇ Traditional database engines read and write a row's data grouped together, resulting in unnecessary columns being read and discarded
- A columnar database stores data grouped by the column it belongs to, instead of the row it belongs to
 - ◇ This is useful in scenarios where you plan to read every row to perform an evaluation - but are only interested in a small subset of the available columns
 - ◇ Such operations perform much less work because it is not reading and then discarding data from columns that it is not interested in
- Because of compression, large datasets can be hosted entirely in memory, instead of needing to be read from a disk
 - ◇ RAM is considerably faster than disk access
 - ◇ RAM is also typically smaller and more expensive than disk space
 - ◇ This does mean that Power BI's usefulness is limited to data sets that can fit in memory

1.A.8 Data Analysis eXpression Language

- DAX Stands for Data Analysis eXpression Language
- Developed by Microsoft as the expression language native to the in-memory columnar database used for information analysis
- Used to add meaning to the data that you are analyzing through formulas, or expressions
- Adds four types of Calculations to our Model
 - ◇ Calculated Columns
 - ◇ Measures
 - ◇ Calculated Tables
 - ◇ Row Filters
- Used everywhere the xVelocity engine is used

Section 1.B

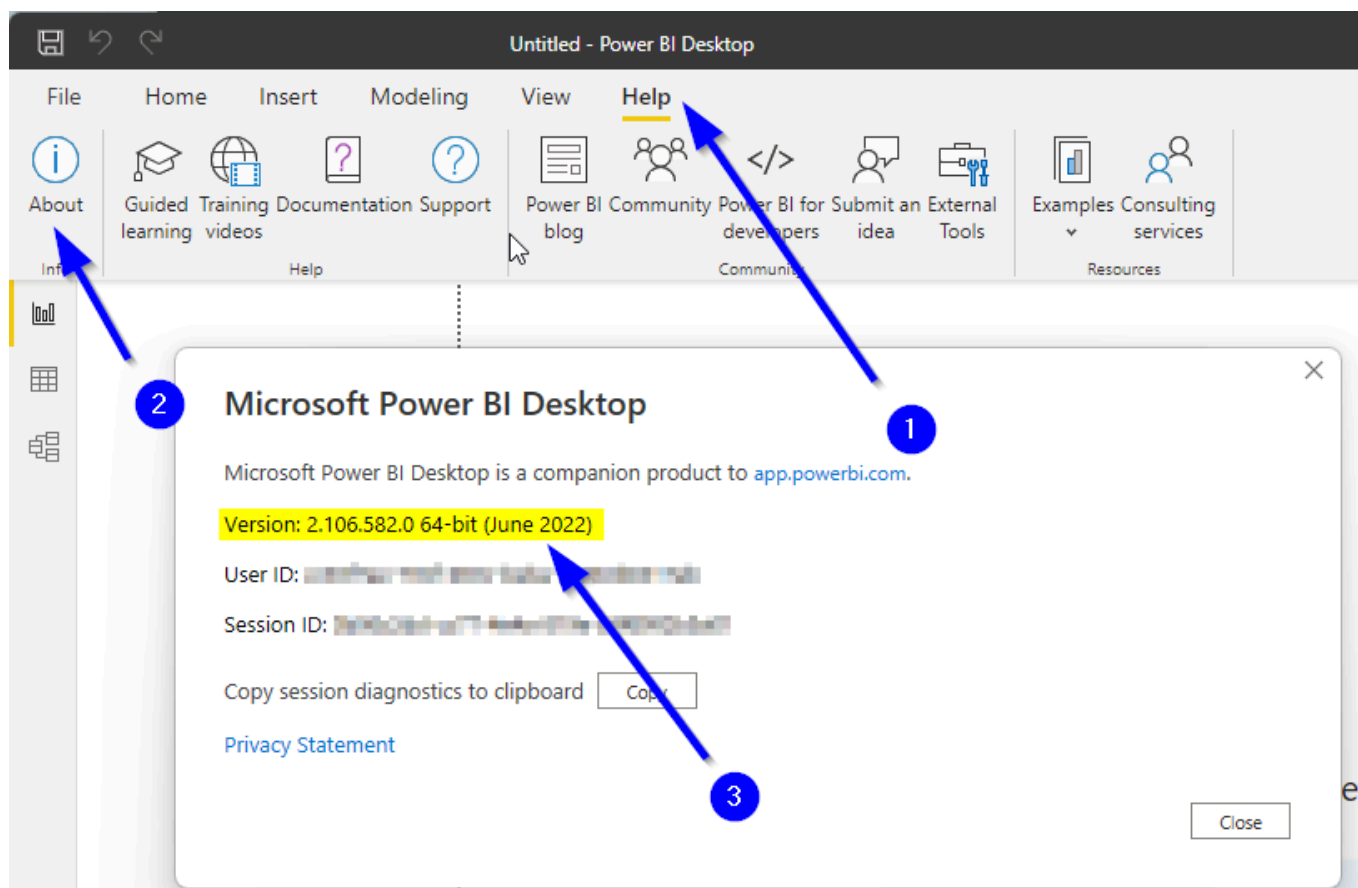
Power BI Tour

1.B.1 Power BI Versioning Strategy

- Microsoft has been very aggressive about releasing updates to Power BI
- The best strategy for sharing files, is to always be on the latest version of Power BI
- If somebody with a newer version of Power BI shares a file with you, you may not be able to open their file until you upgrade

1.B.2 Checking your version of PBI Desktop

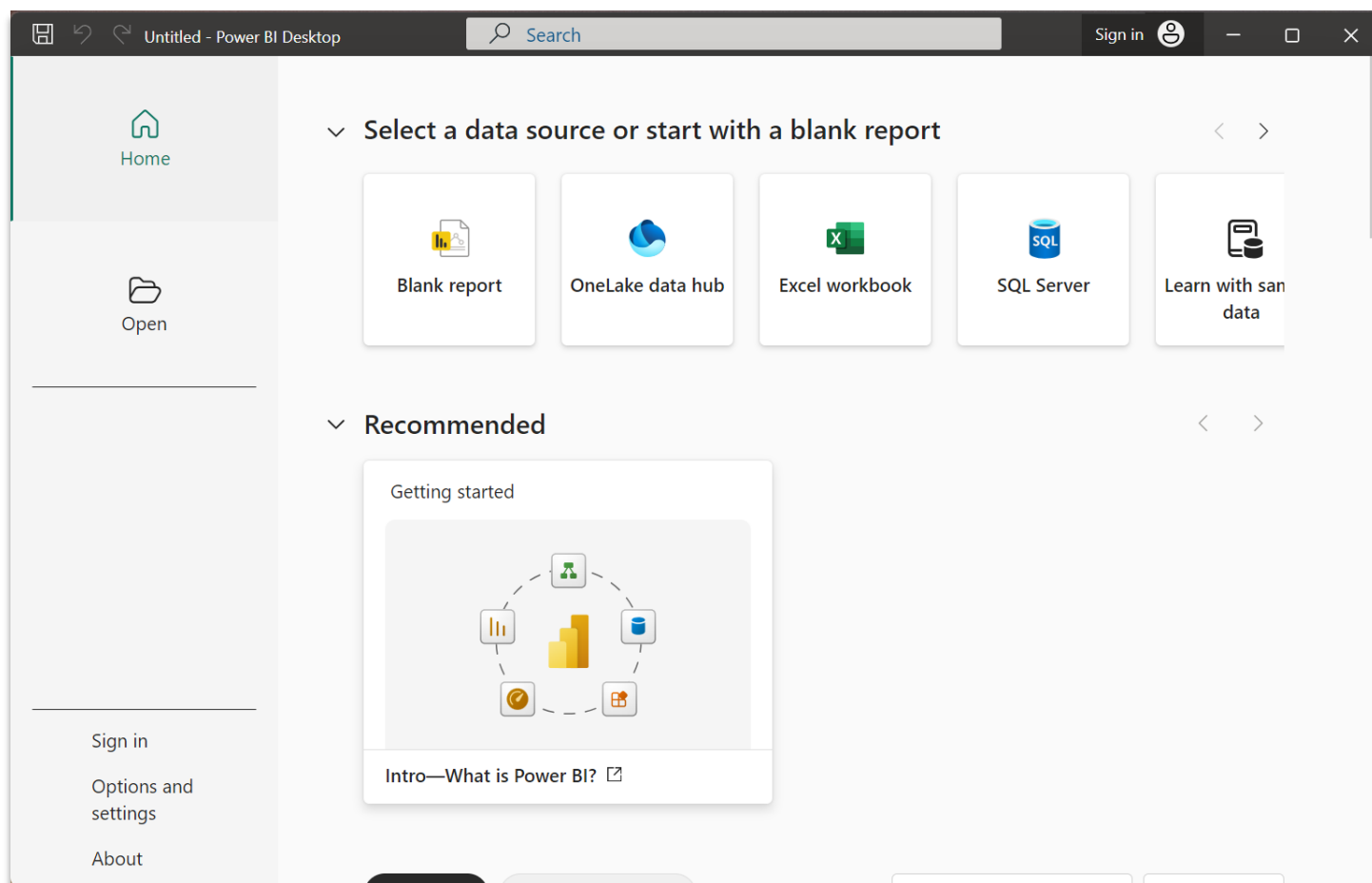
- The following illustration demonstrates how to check your version



1. Navigate to the *Help* Ribbon
2. Select *About*
3. Observe your installed version

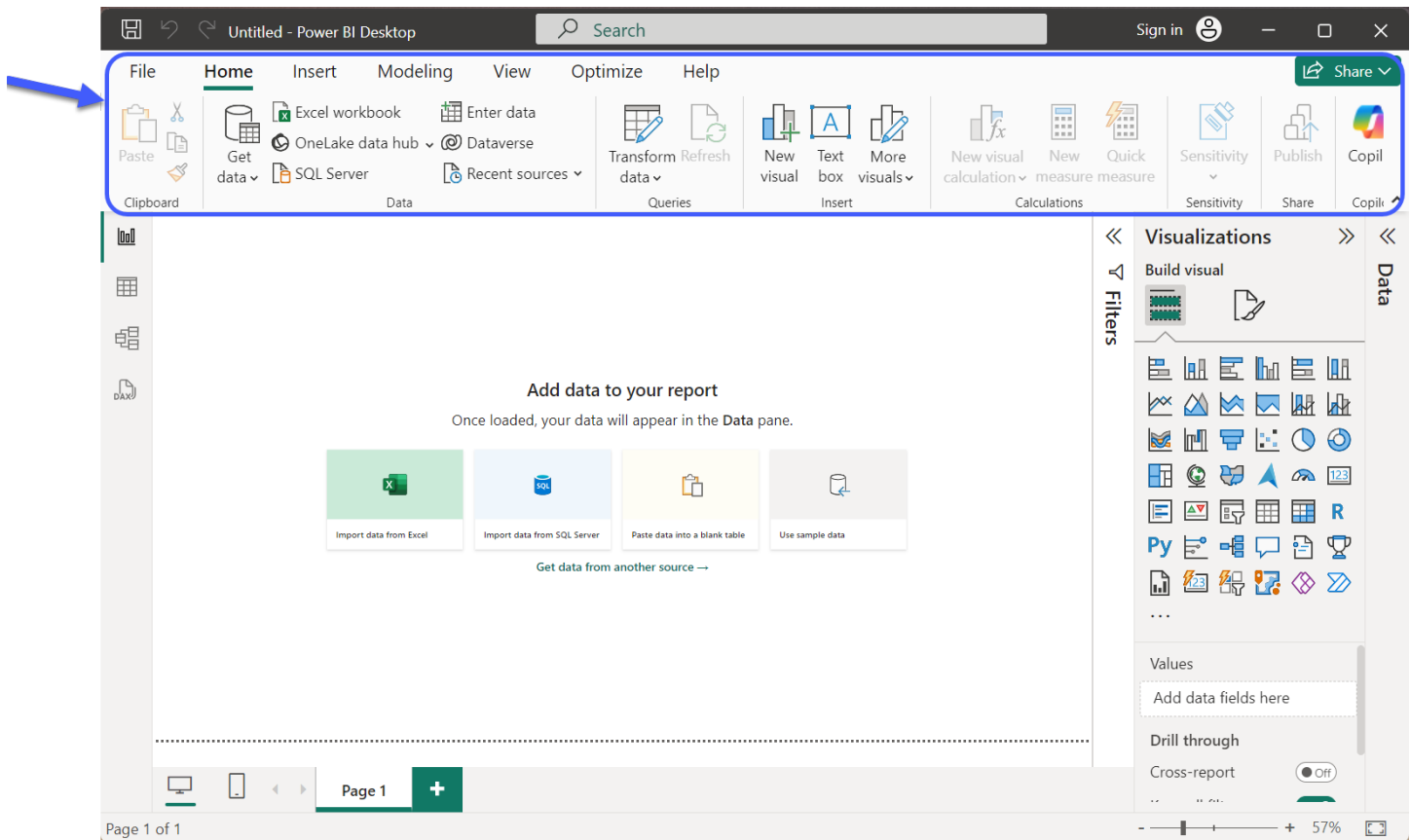
1.B.3 Home Screen

- When the application first loads, a home screen will appear
- This screen can be used to:
 - ◇ Quickly begin a new project from a blank report or specific data source
 - ◇ Quickly open a recent project
 - ◇ Jump to training content under Recommended



1.B.4 The Ribbon

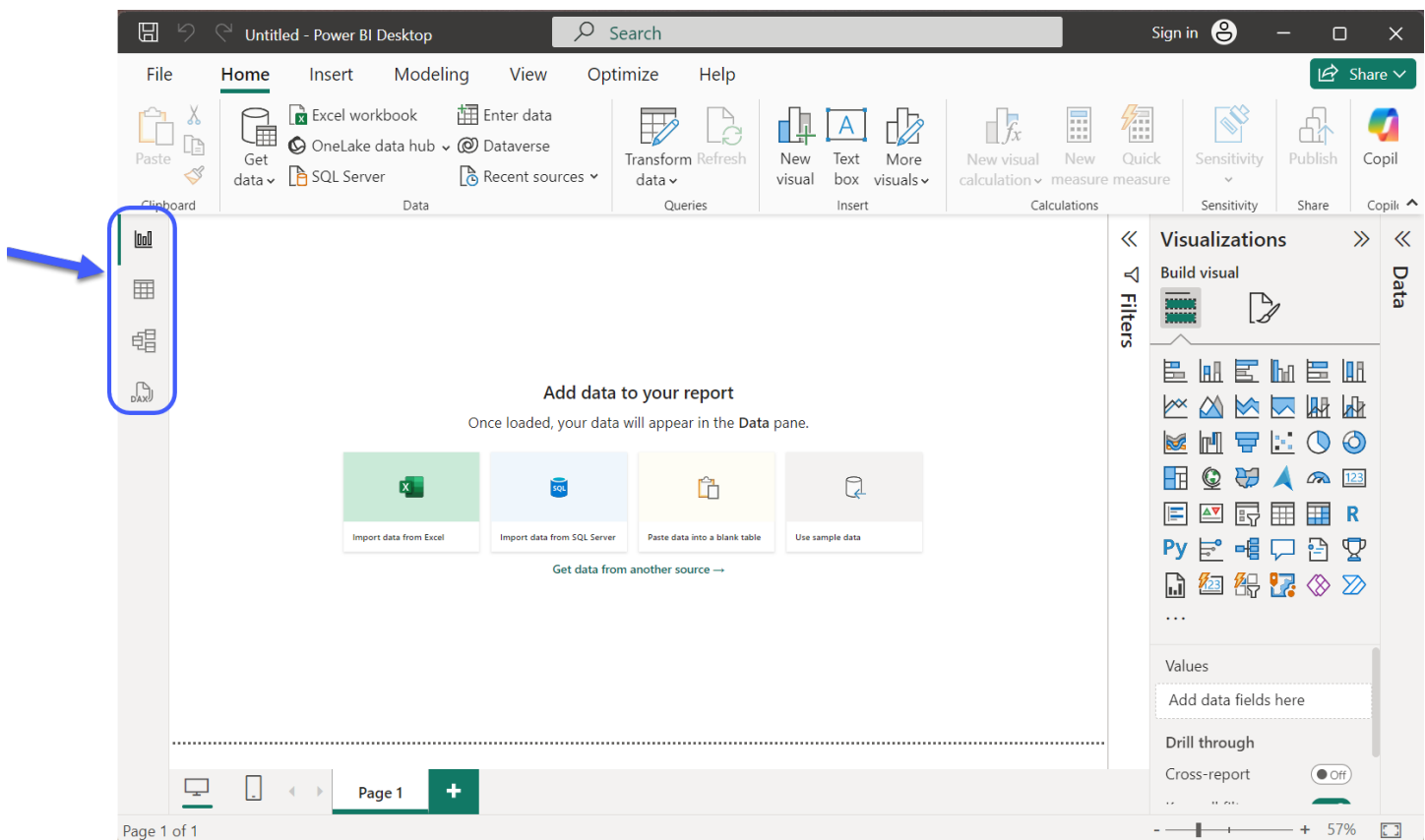
- The top portion of the Desktop interface is called the ribbon



- The tabs on the ribbon are different depending on the selected view, and in some cases what is selected contextually

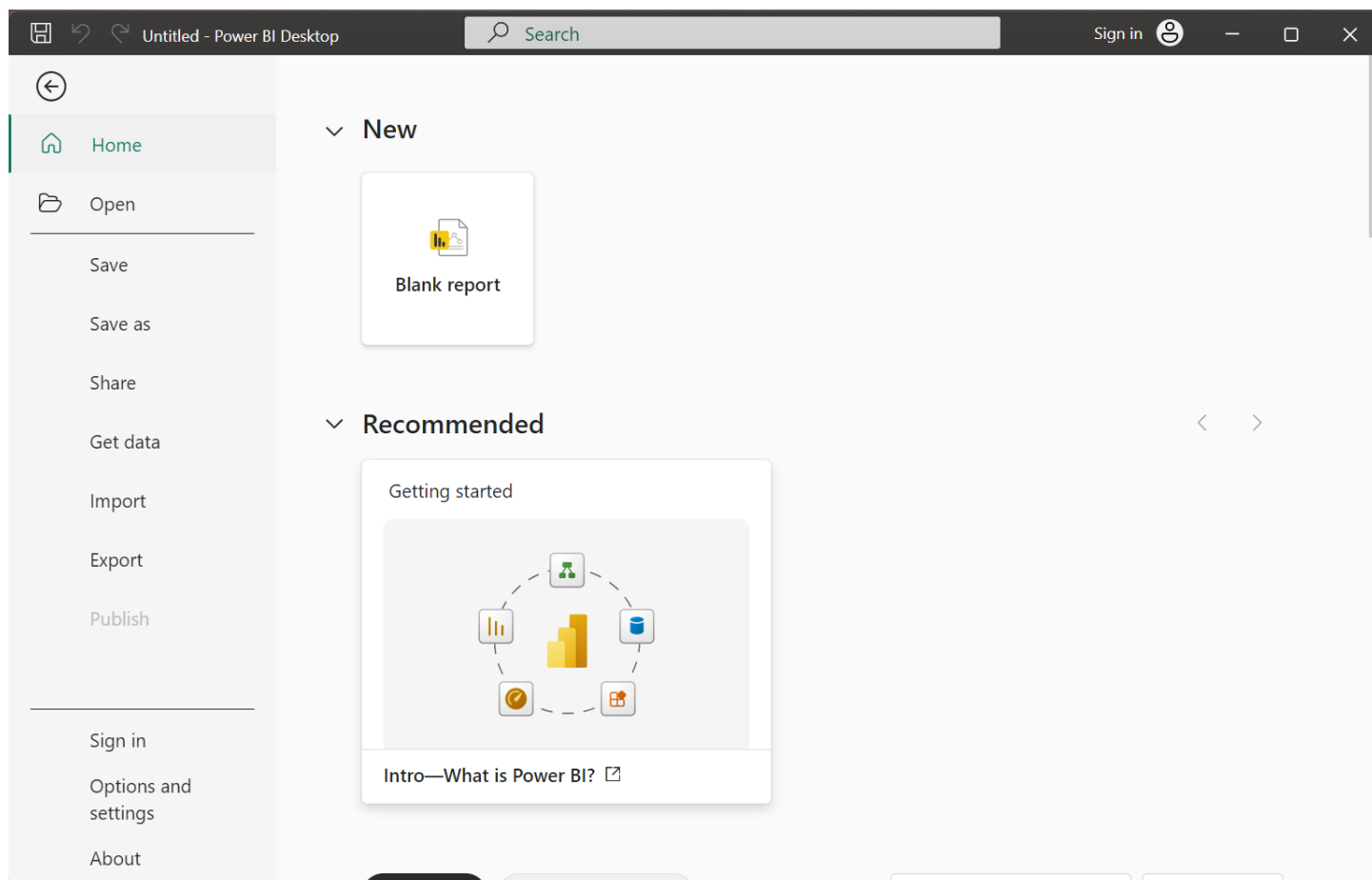
1.B.5 The View Selector

- The View Selector enables the selection of one of four views
 - ◇ Report View: used to design and view visual report elements
 - ◇ Table View: used to view data; some modeling actions can be performed
 - ◇ Model View: used to configure and see the relationship between datasets; some modeling actions can be performed
 - ◇ DAX Query View: allows querying of the semantic model using DAX queries
- The selected view will determine what options are seen in the ribbon and application workspace



1.B.6 The Ribbon: File Tab

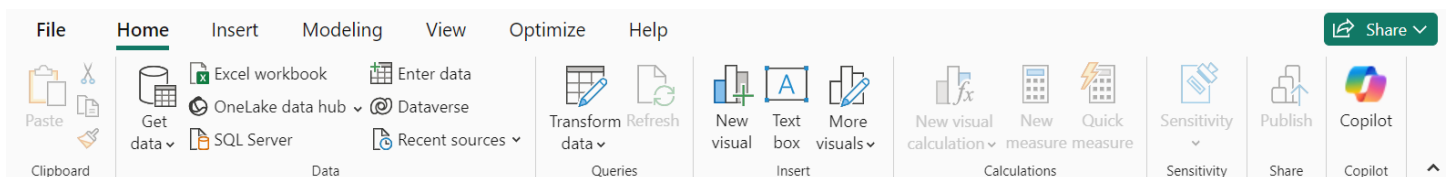
- The File tab will bring up a file menu, where you can manage which project you are working with, and access Options and Settings



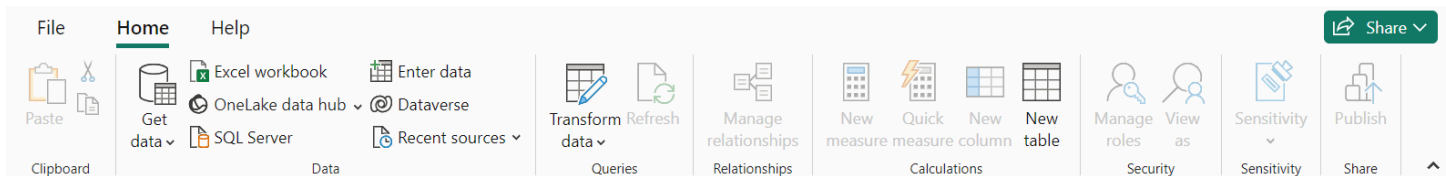
1.B.7 The Ribbon: Home Tab

- The Home tab contains general actions broken down into categories
- Categories and actions available will depend on what View is selected

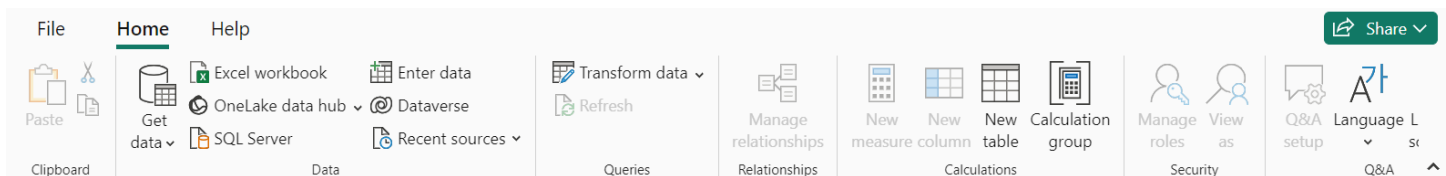
◇ Home ribbon /w Report View



◇ Home ribbon /w Table view

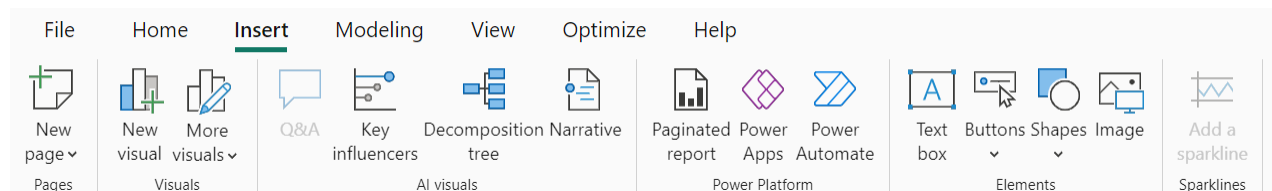


◇ Home ribbon /w Model View



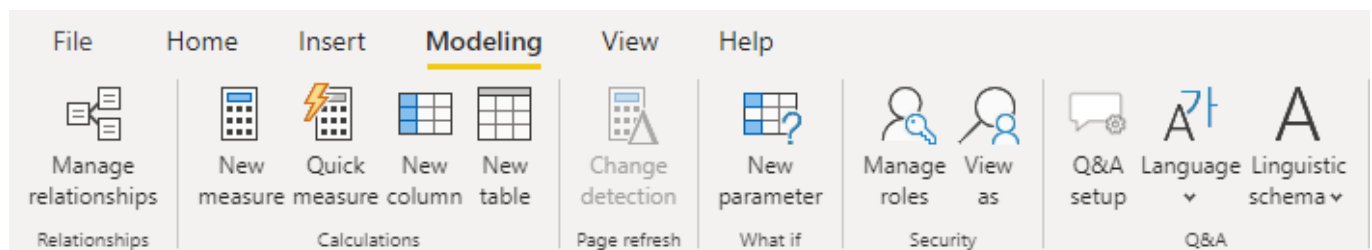
1.B.8 The Ribbon: Insert Tab

- The Insert tab is only available on the Report View and includes various items that can be inserted into a report page



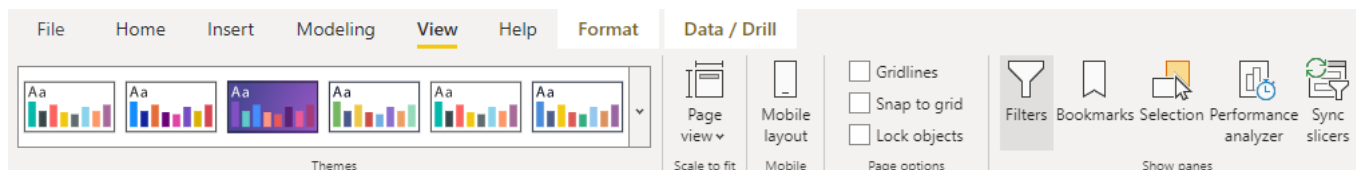
1.B.9 The Ribbon: The Modeling Tab

- The Modeling tab is only available on the Report View
- The Modeling tab offers a variety of shortcuts to alter the model
 - ◇ Most of these actions are accessible contextually from one of the other views using context menus (right clicks) on specific objects



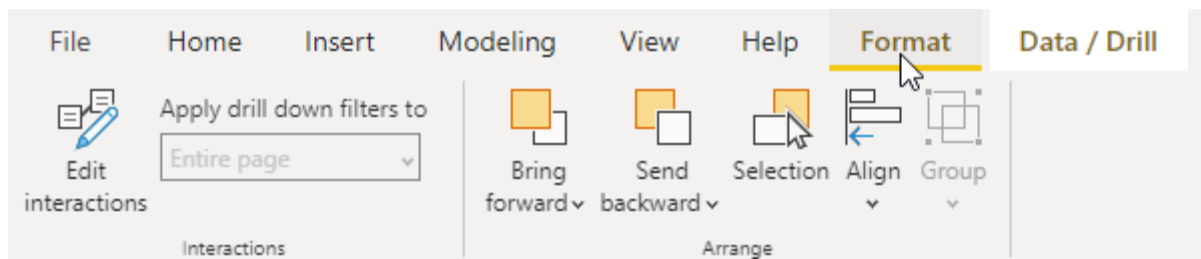
1.B.10 The Ribbon: View Tab

- The View tab is only available on the Report View
- The View tab contains various actions to help control the styling of your reports, or what features are visible in the editor



1.B.11 The Ribbon: Format Tab

- The Format tab is only available on the Report View
- The Format tab becomes visible when a specific report component is selected, and offers actions contextually relevant to the selected component



1.B.12 The Ribbon: Data / Drill Tab

- Only available on the Report View
- Only available when a specific control is selected
- Offers actions contextually relevant to the selected component
- Offers actions to drill down into the data the current component is visualizing

The screenshot shows the 'Data / Drill' tab in the Power BI ribbon. The ribbon includes the following groups and options:

- Visual table**: Visual table, Data point table
- Interactions**: Apply drill down filters to (dropdown: Entire page)
- Drill actions**: Switch to next level, Expand next level, Drill up, Drill down, Drill through
- Data groups**: Groups (dropdown)
- AI**: Find anomalies

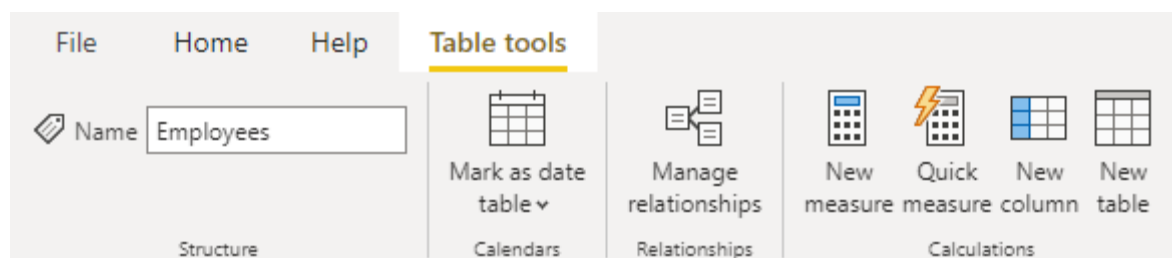
Below the ribbon, a data table is displayed with the following structure:

Year	2019									Total
Quarter	Qtr 2									Total
Month	June									Total
Gender	2	9	16	23	30	Total	Total	Total	Total	
Female	\$38,213	\$21,150	\$24,170	\$17,838	\$35,976	\$137,347	\$137,347	\$137,347	\$137,347	
15 to 20 years	\$10,462	\$4,950	\$5,273	\$3,873	\$7,846	\$32,404	\$32,404	\$32,404	\$32,404	
21 to 25 years	\$13,723	\$8,089	\$8,280	\$6,146	\$14,025	\$50,263	\$50,263	\$50,263	\$50,263	
26 to 30 years	\$10,268	\$5,755	\$7,851	\$5,338	\$9,373	\$38,585	\$38,585	\$38,585	\$38,585	
31 to 35 years	\$3,760	\$2,356	\$2,766	\$2,481	\$4,732	\$16,095	\$16,095	\$16,095	\$16,095	
Male	\$82,071	\$43,552	\$50,122	\$38,810	\$87,584	\$302,139	\$302,139	\$302,139	\$302,139	
15 to 20 years	\$15,029	\$8,124	\$10,434	\$7,877	\$14,708	\$56,172	\$56,172	\$56,172	\$56,172	
21 to 25 years	\$31,511	\$16,596	\$19,549	\$15,372	\$35,901	\$118,929	\$118,929	\$118,929	\$118,929	
26 to 30 years	\$24,771	\$13,122	\$14,478	\$11,173	\$26,712	\$90,256	\$90,256	\$90,256	\$90,256	
31 to 35 years	\$10,760	\$5,710	\$5,661	\$4,388	\$10,263	\$36,782	\$36,782	\$36,782	\$36,782	
Total	\$120,284	\$64,702	\$74,292	\$56,648	\$123,560	\$439,486	\$439,486	\$439,486	\$439,486	

At the bottom right, there is a 'Drill on' dropdown menu set to 'Rows' and several navigation icons.

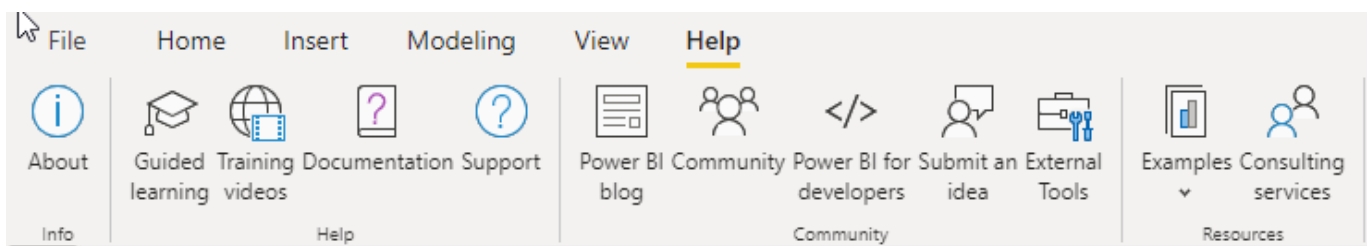
1.B.13 The Ribbon: Table Tools Tab

- The Table Tools tab is available on the Report view *and* Table view, and appears only when a table is selected
- The Table Tools tab provides actions that can be performed on the selected table

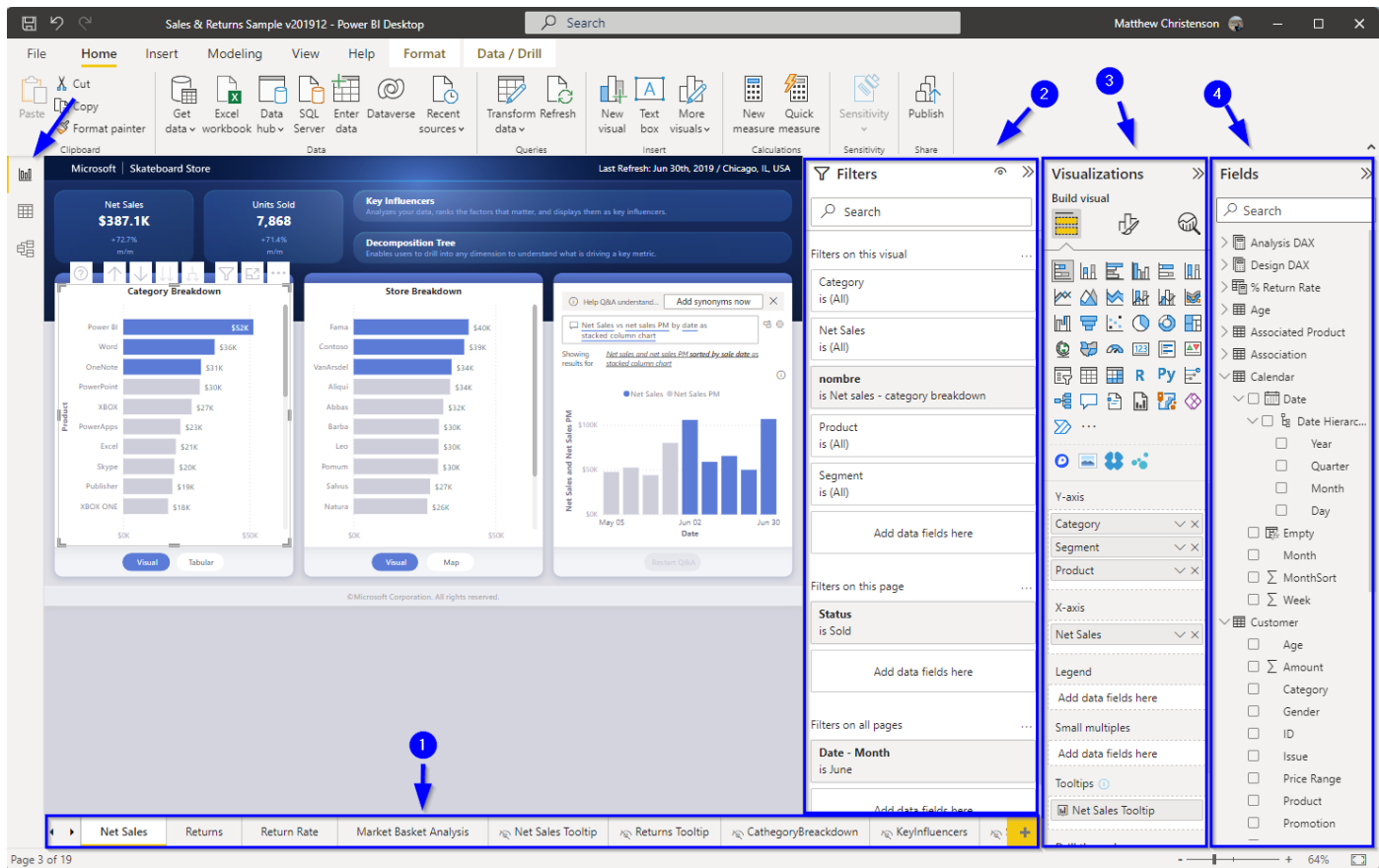


1.B.14 The Ribbon: Help Tab

- The Help tab is available on all three of the Report, Table, and Model views
- The Help tab offers the same options regardless of the view



1.B.15 Report View



1. The Pages area enables you to create multiple report pages and navigate around them
2. The Filters Pane enables you to modify what filters are applied both at the page and visualization level for the selected visualization
3. The Visualizations Pane enables you to add new visualizations to the report surface, or to change the selected visualization's fields and other properties
4. The Fields Pane enables you to select fields that will be included in the selected visualization

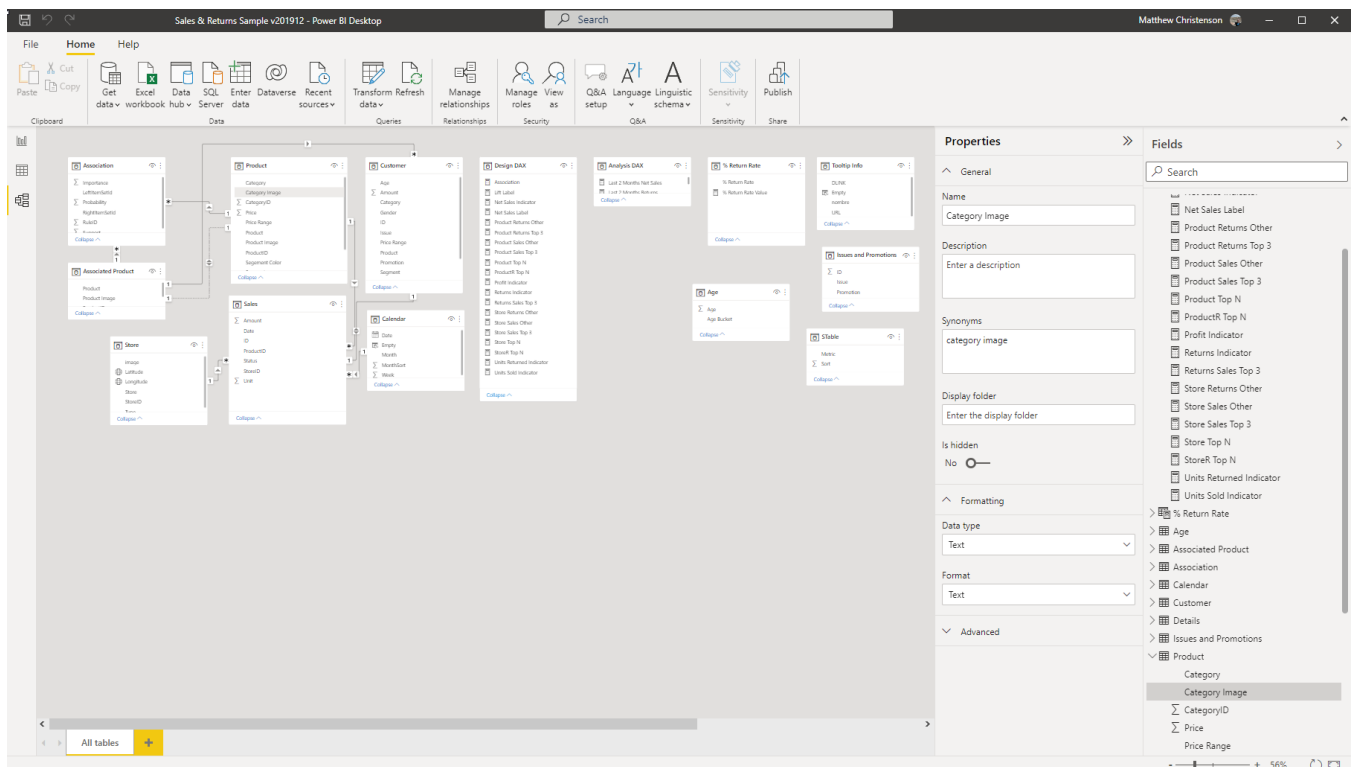
1.B.16 Table View

- The Table View allows you to visualize the raw data in its tables
- You can select columns in the workspace, or select them by expanding a table in the fields pane, and then modify them using the ribbon or context (right click) menus

Table: Sales (29,875 rows) Column: Date (26 distinct values)

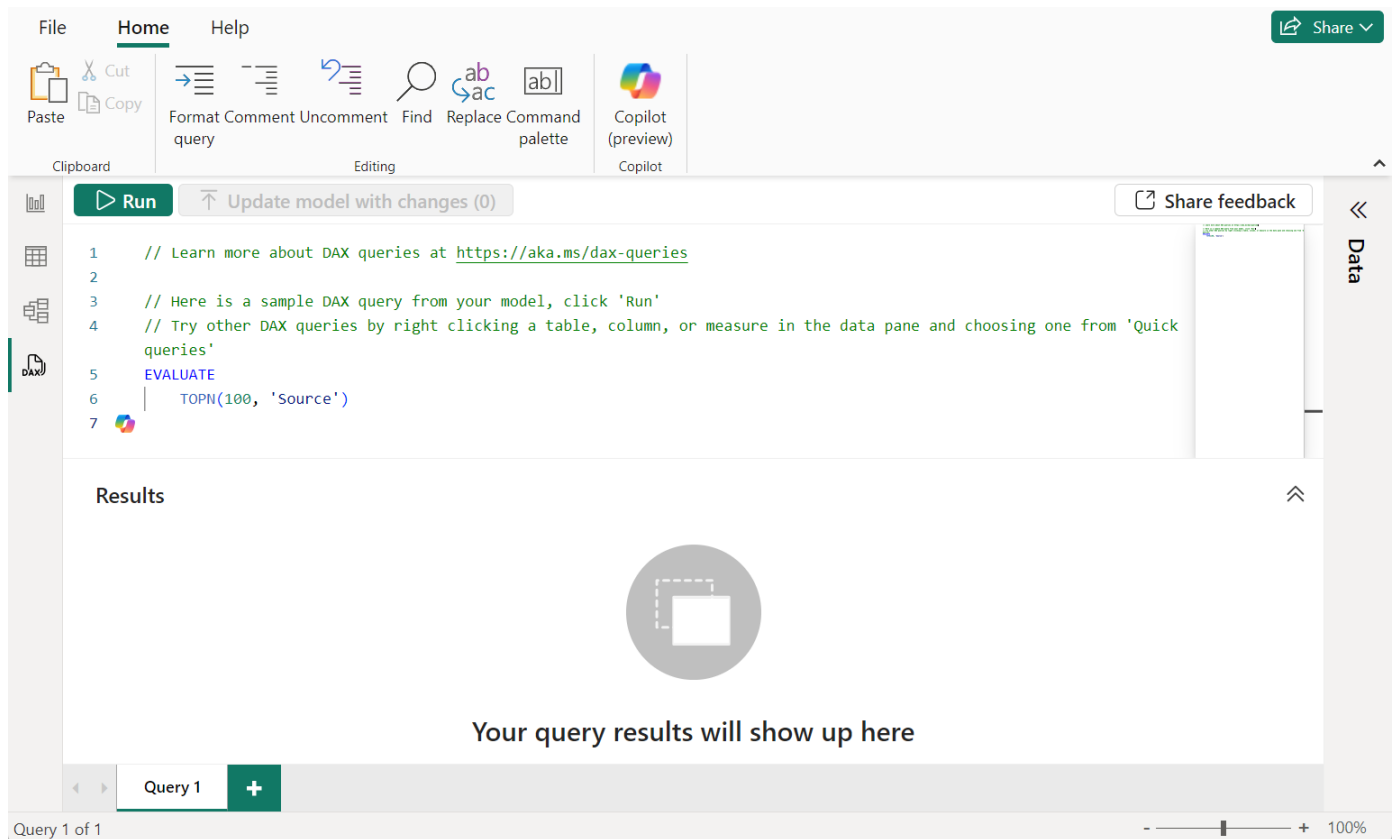
1.B.17 Model View

- The Model View enables you to visualize the relationships between tables
- Additional pages can be added to focus on a smaller subset of tables
 - ◇ This is helpful when you have multiple fact/data tables



1.B.18 DAX Query View

- The DAX Query View is a recently-added feature that allows you to query your Power BI semantic model using queries written in DAX language (similar to querying a database with SQL)
- ◇ DAX queries will be saved to your model, however they will not be visible online when a report is published in the Power BI service



Module 2

Data Modeling with Power BI

Section 2.A

The Data Modeling Process

2.A.1 Modeling Basics

- A data model is the shape that describes relationships within your data, and how each part of the data set relates to the others
- Some modeling is simple
 - ◇ Relational Databases only offer very basic modeling, the shape and datatype of the tables
- Some modeling is more advanced
 - ◇ Advanced Calculations
 - ◇ Explicit Relationship Rules
 - ◇ Attribute Hierarchies
 - ◇ Data Formatting Preferences
- Modeling is a set of guidelines that one might use to shape data
- Modeling is shaping our data in a way that works best for our needs
 - ◇ Different needs mean different models

2.A.2 Modeling Happens in Stages

- Discover the source data
 - ◇ Learn about its existing shape, and understand what values are present, and which will be useful to us
- Bring in the data
 - ◇ Do we have security rights to bring it in?
 - ◇ When in time can we pull in the data?
- Profile the source
 - ◇ Look at the source data and verify that it is what we expected!
- Transform it
 - ◇ Combine Tables
 - ◇ Split Tables
 - ◇ Change Column Names
 - ◇ Change Table Names
 - ◇ Change Data Types
 - * Some models, like Power BI or Data Cubes, allow for more human oriented types like “Percentage”, or “Currency”, where more database centric models would only support “Decimal” or “Float”

2.A.3 Continued Modeling Stages

- Clean up bad data by
 - ◇ Removing it
 - ◇ Fixing it
- Augment the data
 - ◇ Add Calculations
 - * Simple calculations include data from a single row, or from a simple lookup
 - * Advanced calculations consider data from many other rows
 - ◇ Add Attribute Intelligence
 - * Defining Hierarchies
 - * Specifying Date Attributes

2.A.4 Modeling Is Transforming & Describing

- A Source Model is transformed into a Destination Model through a series of steps
- Understanding your source data is essential to the modeling process!
- Understanding an ideal destination model is just as essential!

2.A.5 Source Models

- Understanding various modeling strategies is helpful to understand our starting point
- Source data may come from
 - ◇ Flat Files
 - ◇ OLTP Databases
 - ◇ OLAP Databases
 - ◇ APIs
 - ◇ JSON Files
 - ◇ XML Files

2.A.6 Refresh on Relational Databases

- Relational Databases are made up of multiple tables
- A row in one table may relate to a row in another table
 - ◇ This relationship is frequently recorded as a foreign key
 - * Foreign keys are constraints, not relationships
 - * Relationships are specified with each query, using join statements
 - * The model of relational databases does not have an explicit way to define relationships between tables, but the side effect of foreign keys is “good enough”
- OLTP (Online Transactional Processing) systems:
 - ◇ Process a large number of relatively simple transactions
 - ◇ Enable multi-user access to the same data, while ensuring data integrity
 - ◇ Emphasize very rapid processing, with response times measured in milliseconds
 - ◇ Are highly normalized
- OLAP (Online Analytical Processing) systems:
 - ◇ Structured for reporting in a way that makes data analysis easier
 - ◇ Less normalized than OLTP databases; use star / snowflake schema
 - ◇ OLAP Databases are frequently referred to as Data Warehouses

2.A.7 An Ideal Power BI Model

- An ideal model for Power BI is for data to be organized into a Star, or Snowflake schema
- OLAP databases, also referred to as Data Warehouses, are built using Star Schemas, so these are an ideal data source!
- Frequently OLAP Databases are not available
 - ◇ OLAP Databases require significant planning and design – projects that take months and numerous participants
- Instead, we have two common extremes
 - ◇ Broken into many tables
 - * A large number of tables are involved, having been Normalized to remove any duplicate values
 - * Relationships may form circular loops
 - * Typical of OLTP Data Sources
 - ◇ Flattened into a single table
 - * Relationship data is lost and needs to be inferred based on groupings

Section 2.B

Power BI Modeling

2.B.1 Power BI Models

- In Power BI, a data model is known as a **semantic model**
 - ◇ Uses the xVelocity Engine, essentially the same as models in Excel's Power Pivot, and in SQL Server Analysis Services
 - ◇ This engine is sometimes referenced as “Power Pivot”, which it is called in Excel
- Power BI Modeling happens in the Power BI Desktop Interface
 - ◇ Importing data using Power Query M
 - ◇ Configuring relationships using Model View
 - ◇ Configuring Data Types and preferences
 - ◇ Writing DAX Expressions

2.B.2 Table Modeling

- Ideal Power BI Models take on a similar shape to OLAP databases
 - ◇ Fact tables are frequently referred to as **data tables**
 - ◇ Dimension tables are frequently referred to as **lookup tables**

2.B.3 Star and Snowflake Schema Rules

- It is up to you to enforce star and snowflake schema rules
- Most difficult Data Analysis problems are rooted in a flawed model
- Do not directly relate a data (fact) table with another data table
 - ◇ Data tables should be related only to lookup (dimension) tables
- Do not directly relate a lookup table with another lookup table *unless* it follows snowflake rules
- Relationships between data and lookup tables should be **many** (data records) **to one** (lookup record)

2.B.4 Power BI Model Relationships

- Relationships in Power BI Models are defined explicitly
- Relationships have:
 - ◇ Cardinality (explicitly)
 - ◇ Filter Direction (explicitly)
 - ◇ Existence (implicitly)
- When modeling relationships between tables in Power BI, it is also important to consider the **granularity** of data on each side of the relationship
 - ◇ Granularity refers to the level of detail at which data is stored in a table, e.g. whether the rows of a sales table represent one row per transaction, per day, or per customer
 - ◇ Mismatched granularity can cause incorrect aggregations
 - * For example, joining a daily budget summary table to a transaction-level sales table without accounting for the difference will inflate or distort measures

2.B.5 Data Tables

- Synonymous with fact tables
- Contain “measures,” numeric values that will be summarized
- A record generally describes an event, a thing that happened
- The center table of a star schema

2.B.6 Data Table Naming

- Data tables are best named with plural names representing the thing that happened, such as Sales, Purchases, Shifts, Transfers, Logins, ArticleViews
- Do not prefix tables with Fact in your Power BI semantic model
 - ◇ It ends up looking weird on reports and does not work as well with Natural Language Queries and Q&A

2.B.7 Lookup Tables

- Essentially the same as dimension tables
- Contain descriptive attributes that describe facts
- Are ways by which you would group data to look at fact aggregates
- A record generally describes a noun
- The outer table of a star schema

2.B.8 Lookup Table Naming

- Lookup tables tend to work better if they are named with non-plural names, for example Product, Employee, Currency, and Customer
- Don't prefix these with Dim or Dimension in your Power BI model, as this does not look good in report visuals
- Non-plural names will also work better with the Natural Language Queries feature of Power BI

2.B.9 Columns

- Columns are assigned data types
- Columns can have formatting specified
- Columns can have Sort Order configured to point to a different column
- Columns can have default summarization types assigned
- Columns data can be organized into Groups (categorical) or Bins (numeric)

2.B.10 Column Names

- You would have a tough time getting 10 data engineers in a room and having any 2 of them completely agree when it comes to column naming!
 - ◇ The following guidelines are geared to make your life a little easier when it comes to building powerful visualizations in Power BI
- Avoid ambiguity by adding descriptors to column names
 - ◇ Examples, instead of just “City”:
 - * Supplier City
 - * Order Ship-To City
 - * Customer City
- Remove unnecessary words like Name or Company where they are obvious
 - ◇ Examples:
 - * Customer Name → Customer
 - * Shipping Company → Shipper
- Add words like Amount and Percent to qualify numeric values
 - ◇ Examples:
 - * Discount Amount
 - * Discount Percent

2.B.11 Column Names, continued

- Use spaces where a user would expect to see spaces
- Do not truncate or abbreviate words as a general rule
- Use correct business terminology
 - ◇ Examples:
 - * **Net Sales Amount** is a generally accepted term
 - * **Total Sales Value** is something that may sound fine at first but..
 - * ... Total of what? At what level of grouping?
 - * ... By Sales Value, do you mean the amount of the sale as in revenue, or “sale” as in discount or promotion?
 - * Use **Quick Measures** and templates and suggestions found on-line for ideas
- Use consistent naming across reports and throughout the organization
 - ◇ Include appropriate documentation of report terms used organizationally so that users can be sure they have a clear understanding of what a value represents
- Ask peers for ideas and feedback -- readability is the goal!

2.B.12 Column Datatypes

- Numeric Types
 - ◇ Decimal Number
 - ◇ Fixed Decimal Number
 - ◇ Whole Number
- Date / Time Types
 - ◇ Date / Time
 - ◇ Date
 - ◇ Time
 - ◇ Date / Time / Timezone
 - ◇ Duration
- Text
- True / False
- Binary

2.B.13 Column Numeric Formatting

- All three numeric types can have formatting configuration applied
 - ◇ Can be specified as a currency (multiple currencies)
 - ◇ Can be specified as a *general* number
 - ◇ Can be specified as being a percentage
 - ◇ Can be specified to be displayed with thousands separators (commas)
 - ◇ Can be specified to be displayed in scientific notation
 - ◇ Can be specified to be rounded to a specific decimal place

2.B.14 Date / Time column Formatting

- Date / Time columns can be assigned preferred text formatting rules

Date formats

*3/14/2001 (Short Date)

*Wednesday, March 14, 2001 (Long Date)

Wednesday, March 14, 2001 (dddd, mmmm d, yyyy)

March 14, 2001 (mmmm d, yyyy)

Wednesday, 14 March, 2001 (dddd, d mmmm, yyyy)

14 March, 2001 (d mmmm, yyyy)

3/14/2001 (m/d/yyyy)

3/14/01 (m/d/yy)

03/14/01 (mm/dd/yy)

03/14/2001 (mm/dd/yyyy)

01/03/14 (yy/mm/dd)

2001-03-14 (yyyy-mm-dd)

14-Mar-01 (dd-mmm-yy)

14/03/2001 (dd/mm/yyyy)

March 2001 (mmmm yyyy)

2001-03 (yyyy-mm)

March 14 (mmmm d)

01 (yy)

2001 (yyyy)

2.B.15 Column Data Categories

- Data categorization can enable specialized visualization behavior
- Options include:
 - ◇ Uncategorized
 - ◇ Geography Classifications:
 - * Address
 - * Place
 - * City
 - * County
 - * State or Province
 - * Postal Code
 - * Country
 - * Continent
 - * Latitude
 - * Longitude
 - ◇ Hyperlink Classifications:
 - * Web URL
 - * Image URL
 - ◇ Barcode

2.B.16 Column Sort Orders

- Columns can have a Sort By column defined
- Examples of this being useful include:
 - ◇ Sorting a Display Name that contains a First and Last name, by a separate Last Name column
 - ◇ Sorting a Month Name column by a Month Number column so that January appears before July (or Day Name by Day-of-Week Number)
 - ◇ Sorting discretized data like Infant, Child, Teen, Young Adult, Adult, Senior based on their logical sort order

2.B.17 Changing Default Summarization

- The default summarization for numeric columns is Sum
- This works great for quantities, amounts, etc.
- Measures like Popularity that contains a 0-5 value would not make sense to sum
 - ◇ An average would be a better aggregation for this column
- This controls the implicit measures created for each column when selecting them for a visualization
 - ◇ You can override this within the visualization
- You can also add more measures explicitly to the model
 - ◇ A Year column may not have an obvious default, but two custom measures of might be created using Minimum and Maximum with better names:
 - * First Year
 - * Most Recent Year
 - ◇ We will cover measures in detail when we cover DAX

2.B.18 Navigational Hierarchies

- Navigational **hierarchies** can be created that specify a drill-down order to be added to visualizations
- Some hierarchies are obvious and follow natural parent child relationships
 - ◇ Year > Quarter > Month > Day
 - ◇ Country > Region > County > City
- Some hierarchies do not have as natural relationships; these are frequently ordered by those with the fewest options to the most options, but ultimately depends on your visualization objectives
 - ◇ Gender > Marital Status > Age Bracket
 - ◇ Category > Size > Color

Section 2.C

Data Transformation

2.C.1 Transformation Pipelines

- When building a new model, transformation happens in stages
- In Power BI, transformation is done with:
 - ◇ Power Query using M language
 - ◇ Configuration through the graphical user interface (GUI)
 - ◇ Adding DAX Expressions

2.C.2 Transformation Order

- The order in which transformations take place might vary from case to case, but generally you'll want to do things in this order:
 1. Bring in raw data to a staging table
 2. Remove the columns you won't need
 3. Remove or repair data that does not conform to column types
 4. Remove data that is out of the scope of analysis
 5. Define the column types
 6. Define Column Names
 7. Fill unknowns / nulls
 8. Add row level calculations as desired
 - a. Add row level totals
 - b. Add formatted variations
 - c. Pulling parent data into child tables
 - d. Pulling child aggregates into parent tables

2.C.3 Cleaning Column Data

- Address correction and correcting city spellings
- Data abnormalities:
 - ◇ How many years have you been teaching?
 - * 4
 - * 16
 - * 12 year
 - * 2011
- Fixing implausible data or obvious typos:
 - ◇ Age?
 - * 999
 - ◇ Gender
 - * Malf
 - ◇ However, always be very careful about making assumptions with data!

2.C.4 Filling Unknowns with Lookups

- Demographic information
 - ◇ Average income for a city, zip code, or region
 - ◇ School district associated with zip code
- Economic data
 - ◇ Interest rates over a specified period
 - ◇ Unemployment rates or job growth over a period
- Weather data
 - ◇ Annual rainfall in each city?
- Geological detail
 - ◇ How many feet above sea level?
 - ◇ What is the population of this region?

2.C.5 Augmenting Column Data

- Adding formatted variations
 - ◇ You might add a “Display Name” column from a First and Last Name
- Pulling parent data into children
 - ◇ You may want the Products Table to contain the Category Name instead of referencing a separate Category table
- Pulling children aggregates into parents
 - ◇ You may want an **Order** table to have an **[Order Total]** field
 - ◇ You may want an **Employee** Table to have a **[Number of Sales Regions]** field
- Using discretized fields
 - ◇ You might opt to transform birthdate into one of the following:
 - * Free Child (3 and younger)
 - * Child (4-12)
 - * Teen (13-18)
 - * Young Adult (19-24)
 - * Adult (25+)
 - * Senior (60+)

Module 3

Power Query GUI Experience

Section 3.A

Power Query

3.A.1 Power Query

- Power Query is used to bring data into Power BI
 - ◇ Power Query was designed to be approachable to users who do not have any coding experience
 - ◇ It was designed to support a diverse set of data sources
 - ◇ There are over 700 transformations
- Power Query Editor is a graphical user interface (GUI)
 - ◇ “Wizards” and dialogs can be used to bring data in from a data source
 - ◇ Various tools on the ribbon can be used to perform basic transformations to data as it is brought in, before the data is made available to the Power BI Model
- Power Query Formula Language M is the underlying code that describes where data comes from and what happens to it
- The Advanced Editor can be used to write your own M expressions

3.A.2 Power Query Steps

- Queries are broken down into steps
 - ◇ Each step has a name and performs a transformation
 - ◇ Some steps, such as changing Data Types, can operate on multiple columns at the same time
- Together the steps form a simple transformation pipeline

3.A.3 Power Query Automates Manual Processes

- Excel has often been used to clean and filter data prior to importing it into reporting solutions
 - ◇ When presented with a new Excel file and a new set of data, users had to manually repeat the steps they performed to clean and prepare data
- Power Query serves to automate this kind of data preparation work
- Power Query can be thought of as a macro language
- When your source data changes, you can run Refresh Data to fetch updates and re-apply each of the query steps again

3.A.4 Query Folding

- Power Query utilizes a technique known as query folding, which pushes a lot of the transformation logic downstream for Get Data adapters that support it
- Data sources like Microsoft SQL Server have work like filtering, exclusion of columns, joining, and grouping pushed into the SQL query
 - ◇ This means much less data gets transferred into Power BI
 - ◇ This allows database engines like MySQL or SQL Server to utilize indexes to perform expensive operations like filtering or ordering data more effectively

3.A.5 Power Query Uses

- Power Query is great for
 - ◇ Extracting Data from a variety of sources
 - ◇ Shaping data in a variety of ways
 - ◇ Merging data
 - ◇ Transforming data
 - ◇ Cleaning data
 - ◇ Calculating or manipulating values
- Not the best tool for
 - ◇ Auditing or logging
 - ◇ Strict data type enforcement
 - ◇ Non-linear data prep
 - ◇ Optimized for ease-of-use over speed
 - ◇ Starts to break down when the number of steps exceeds 20
 - ◇ Performance or Scalability
- Other products might be considered for more complex ETL pipelines
 - ◇ Microsoft SQL Server Integration Services is one popular option

3.A.6 Products that use Power Query

- Power Query is used in a variety of Microsoft products

<https://docs.microsoft.com/en-us/power-query/power-query-what-is-power-query>

Product	M engine ¹	Power Query Desktop ²	Power Query Online ³	Dataflows ⁴
Excel for Windows	Yes	Yes	No	No
Excel for Mac	Yes	No	No	No
Power BI	Yes	Yes	Yes	Yes
Power Apps	Yes	No	Yes	Yes
Power Automate	Yes	No	Yes	No
Power BI Report Server	Yes	Yes	No	No
Azure Data Factory	Yes	No	Yes	Yes
SQL Server Integration Services	Yes	No	No	No
SQL Server Analysis Services	Yes	Yes	No	No
Dynamics 365 Customer Insights	Yes	No	Yes	Yes

¹ M engine	The underlying query execution engine that runs queries expressed in the Power Query formula language ("M").
² Power Query Desktop	The Power Query experience found in desktop applications.
³ Power Query Online	The Power Query experience found in web browser applications.
⁴ Dataflows	Power Query as a service that runs in the cloud and is product-agnostic. The stored result can be used in other applications as services.

3.A.7 Dataflows

- We will be working with Power Query within the context of Power BI Desktop, where the tooling is tightly coupled with a project
- A dataflow decouples the data transformation layer from the modeling and visualization layer in a Power BI solution
- Dataflows utilize Power Query in an application-agnostic way, on the cloud
- Power BI Desktop can consume cloud hosted Dataflows

<https://docs.microsoft.com/en-us/power-query/dataflows/overview-dataflows-across-power-platform-dynamics-365>

3.A.8 Power Query Formula Language M

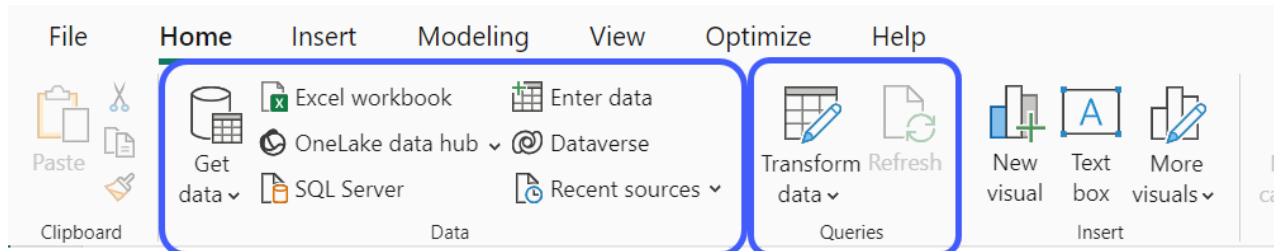
- Much of what Power Query accomplishes can be performed by using Power Query Editor GUI
- As you interact with the Power Query Editor, it is automatically writing queries using Power Query Expression Language (M)
 - ◇ We will refer to it as M, or M Language throughout this course
 - ◇ When you review documentation and external learning content, be aware that Microsoft seems to be very inconsistent with how they work M into their documentation
- Understanding how to write your own M Language queries will enable you to perform more advanced tasks with Power Query
 - ◇ Often you'll start with queries written by Power Query Editor's GUI, and then use the advanced editor to modify them to perform more advanced tasks
- This module focuses on learning to navigate around Power Query Editor, and using the GUI to get data and perform common transformations
 - ◇ We will cover M Language in more depth later

Section 3.B

Power Query Editor Tour

3.B.1 Getting to the Power Query Editor

- The “Data” and “Queries” sections of the Home ribbon contain shortcuts related to Power Query

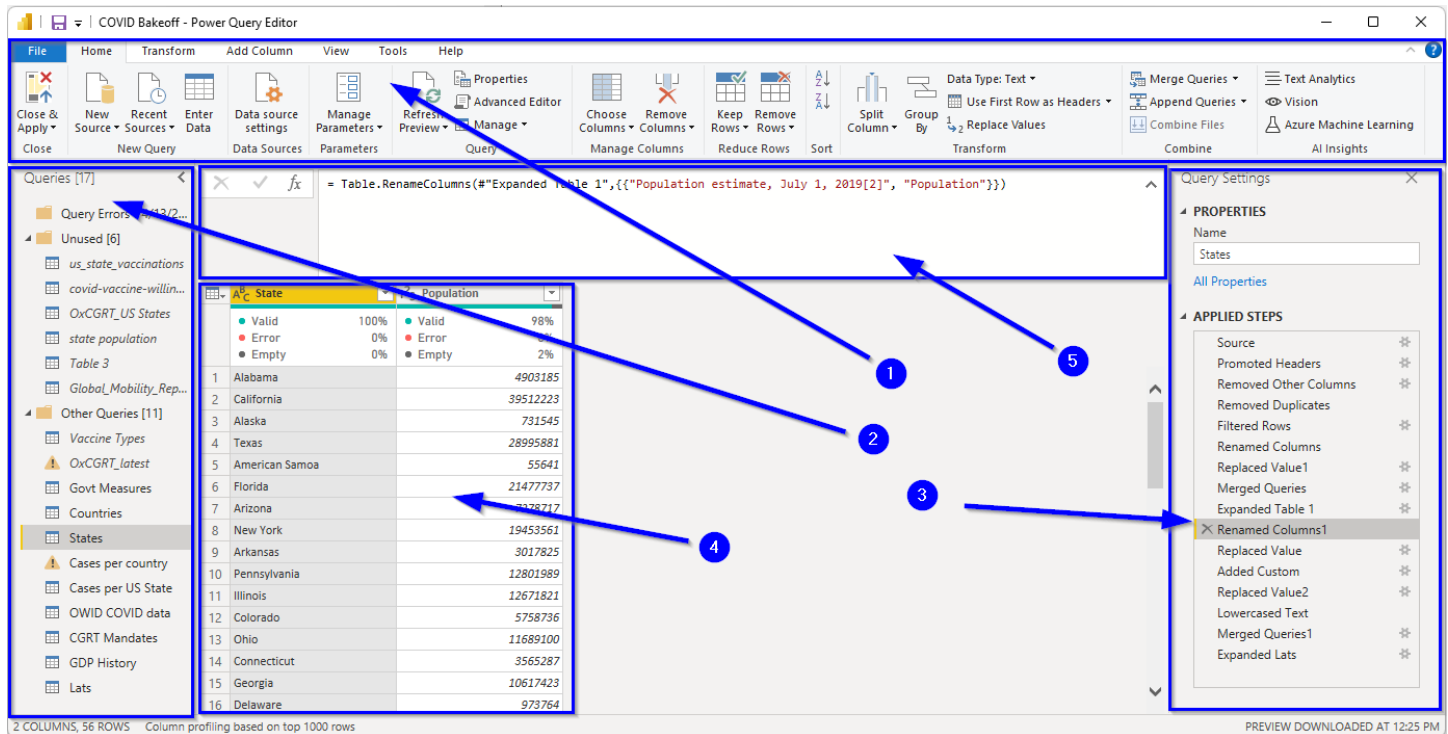


- **Get Data** allows you to bring in a wide variety of data sources
- **Transform Data** launches Power Query

3.B.2 Supported Data Sources

- Power Query supports a wide variety of data sources including:
 - ◇ Databases
 - * Local, on-premises or cloud accessible
 - * SQL Server (Microsoft), MySQL, Oracle, PostgreSQL, many more
 - ◇ Files
 - * Stored locally or in SharePoint or OneDrive
 - * Excel, Text, CSV, XML, JSON, etc.
 - * May import a Folder to quickly bring in a full set of files
 - ◇ Web
 - * Url Sources
 - * OData feeds
 - ◇ Microsoft Fabric and Azure
 - * Dataflows, Warehouses, Lakehouses
 - * Cloud Databases, Blob Storage
 - ◇ Online Services (SaaS sources)
 - * Salesforce, Google Analytics, Adobe, and others
 - ◇ Power Platform
 - * Dataflows, Dataverse, Data Service (Legacy)

3.B.3 Power Query Editor: Image Tour



1. The Ribbon Pane contains many useful shortcuts to contextually bring in or transform data
1. The Queries List Pane has a list of all queries already in your project
 - * Queries can be organized into folders if you have a lot of them
2. The Query Settings Pane has a list of Steps that make up your Query
 - * Steps can be reordered or removed in this pane
 - * A gear appears next to some steps that enable you to modify it using a custom graphical dialog
3. The body region allows you to preview the data after the selected step has been applied
 - * You can also right click on columns to see a context menu, double click to rename, and drag to reorder
4. The Expression Editor allows you to edit the code based expression behind the selected step
 - * You can expand this from a single line to multiple lines to make editing easier

3.B.4 Demonstration: Power Query Editor Tour

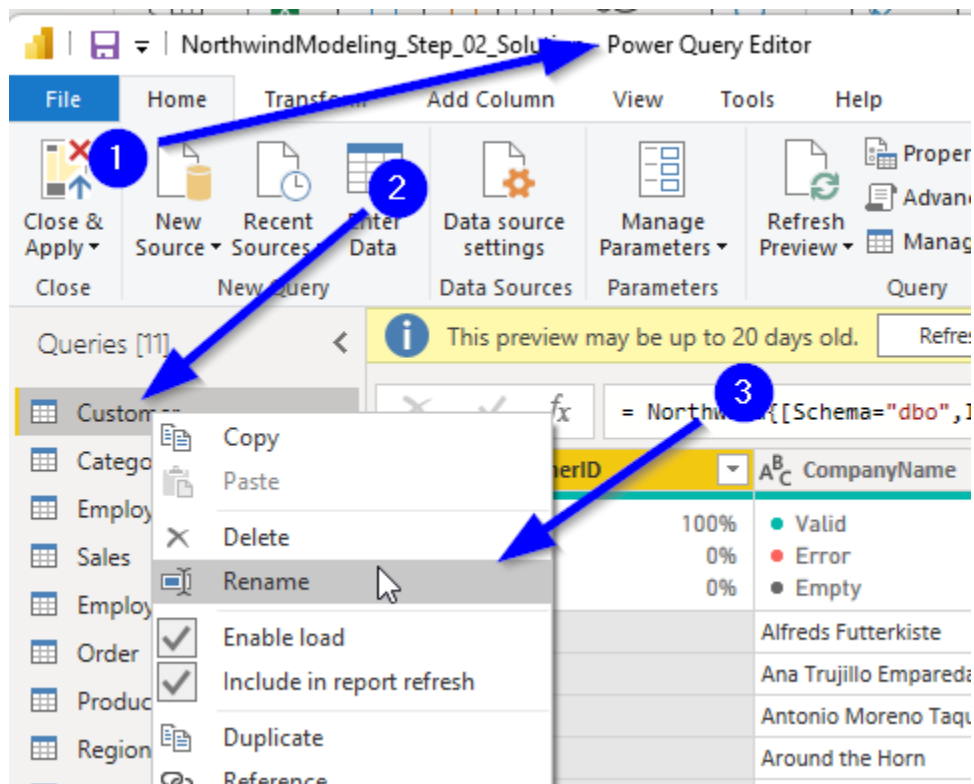
- Let's pull in some data and shape it!

- ◇ Getting Data
- ◇ Removing Columns
- ◇ Renaming Columns
- ◇ Changing Column Types
- ◇ Formatting Columns
- ◇ Creating Calculated Fields

Section 3.C

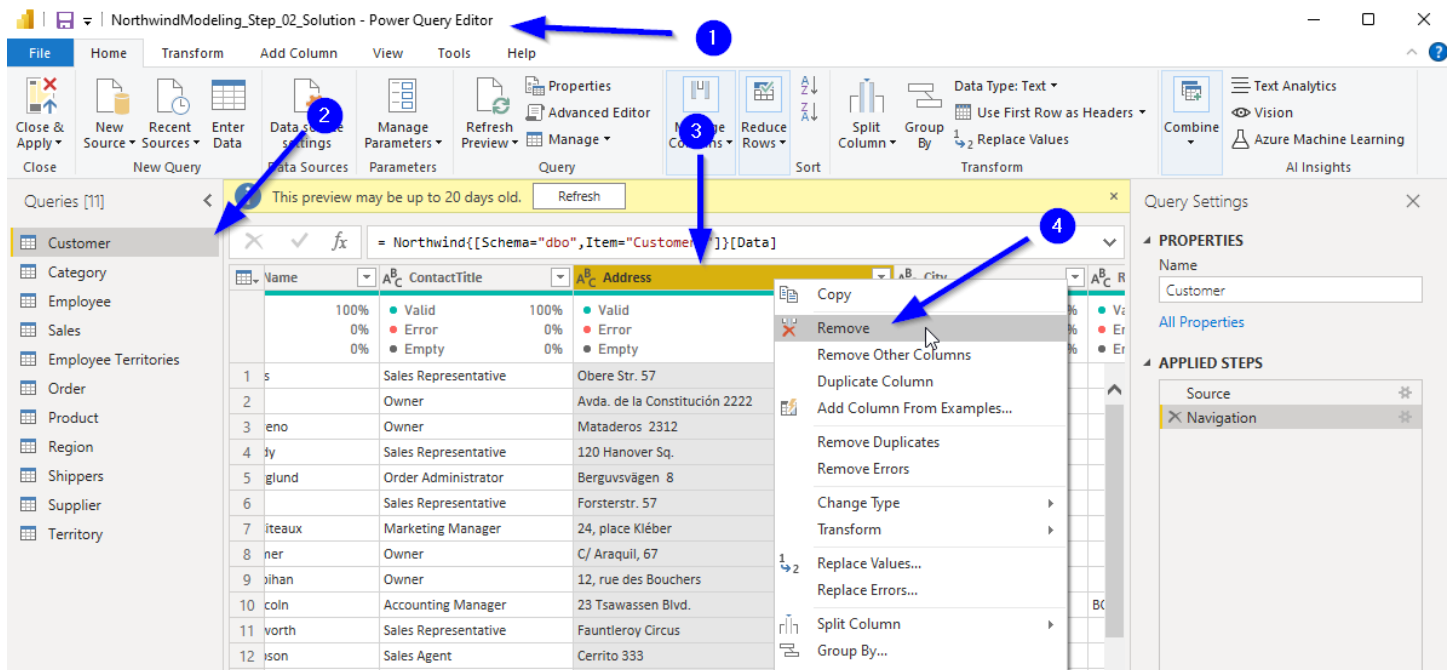
Simple Transformations

3.C.1 Changing a Query/Table Name



1. Using Power Query Editor
2. Select the Query (table) that you want to change, and Right Click to open the context menu
3. Select Rename

3.C.2 Removing Columns



1. Using Power Query Editor
2. Select the Query that contains the Column you want to remove
3. Select and then right click on the column you want to remove, to open the context menu
4. Select Remove

3.C.3 Renaming Columns

NorthwindModeling_Step_04_Solution - Power Query Editor

File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Sources Enter Data Data source settings Manage Parameters Refresh Preview Properties Advanced Editor Manage Choose Columns Remove Columns Keep Rows Remove Rows Sort

Queries [11]

Customer

Category Employee Sales Employee Territories Order Product Region Shippers Supplier Territory

This preview may be up to 20 days old. Refresh

= Table.RemoveColumns(dbo_Customers,{"Address", "Phone", "Fax"})

	CustomerID	CompanyName	ContactName
1	ALFKI	Alfreds Futter	
2	ANATR	Ana Trujillo Er	
3	ANTON	Antonio More	
4	AROUT	Around the H	
5	BERGS	Berglunds sna	
6	BLAUS	Blauer See De	
7	BONP	Blondesdsl p	
8	BOLID	Bólido Comida	
9	BONAP	Bon app'	
10	BOTTM	Bottom-Dollar	
11	BSBEV	B's Beverages	
12	CACTU	Cactus Comida	
13	CENTC	Centro comer	
14	CHOPS	Chop-suey Ch	
15	COMMI	Comércio Min	
16	CONSH	Consolidated	
17	DRACD	Drachenhut D	
18	DUMON	Du monde ent	
19	EASTC	Eastern Conne	

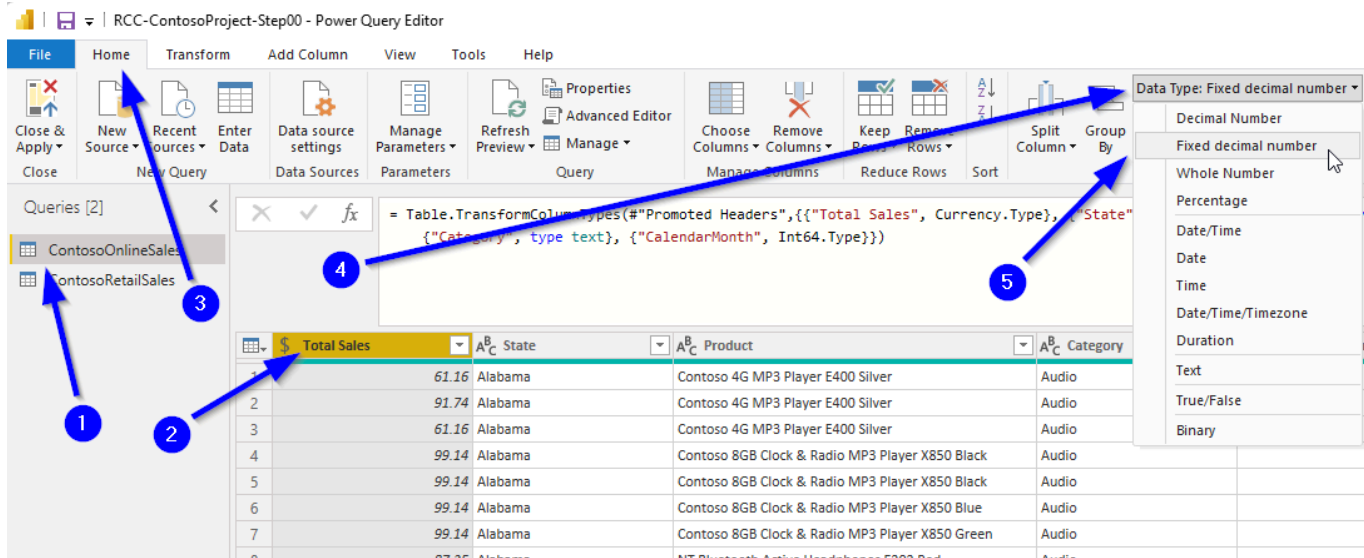
Copy
Remove
Remove Other Columns
Duplicate Column
Add Column From Examples...
Remove Duplicates
Remove Errors
Change Type
Transform
Replace Values...
Replace Errors...
Split Column
Group By...
Fill
Unpivot Columns
Unpivot Other Columns
Unpivot Only Selected Columns
Rename...
Move

1. Using Power Query Editor
2. Select the Query that contains the Column that you wish to rename
3. Select and then right click on the column you want to rename, to open the context menu
4. Select Rename

3.C.4 Changing a Column Data Type

- Selecting appropriate data types for your columns is important!
- Data types will determine what transformations will work on values
 - ◇ Numeric aggregations like Sum or Average will only work on numeric types
 - ◇ Transformations like Split Column will only work on text columns
 - ◇ Sometimes the user interface will hide or deactivate transformations that are not relevant to the selected column's current data type
- Some data sources, such as SQL Server, carry information over about the correct data type
 - ◇ There will be little work to do here
 - ◇ It is still worth reviewing, as some databases may surprisingly use unexpected types
- Data sources such as text files are just guessing data types
 - ◇ These should be reviewed column by column to confirm the type is correct

3.C.5 Changing a Column Data Type: Illustrated



5. Select the Query
6. Select the Column
7. Select the Home Ribbon
8. In the Transformation Region, select the Data Type dropdown
9. Select the desired Data Type

3.C.6 Adding Calculated Fields

The screenshot shows the Power Query Editor interface. The 'Add Column' tab is selected in the ribbon. The 'Custom Column' dialog box is open, showing the formula for creating a new column named 'Employee Description'.

Numbered steps:

1. Click the 'Add Column' tab in the ribbon.
2. Click the 'Custom Column' button in the 'Add Column' group.
3. Click the 'Custom Column' button in the 'Add Column' group.
4. Click the 'Employee' query in the 'Queries' list on the left.
5. Click the 'Custom Column' button in the 'Add Column' group.
6. Click the 'OK' button in the 'Custom Column' dialog box.

Custom Column Dialog Box Details:

- New column name:** Employee Description
- Custom column formula:** `= [Employee Title of Courtesy] & " " & [Employee First Name] & " " & [Employee Last Name] & " (" & [Employee Title] & " )"`
- Available columns:** EmployeeID, Employee Last Name, Employee First Name, Employee Title, Employee Title of Courtesy, Employee Birth Date, Employee Hire Date.
- Status:** No syntax errors have been detected.

EmployeeID	Employee Last Name	Employee First Name	Employee Title	Employee Title of Courtesy	Employee Birth Date
1	Davolio				
2	Fuller				
3	Leverling				
4	Peacock				
5	Buchanan				
6	Suyama				
7	King				
8	Callahan				
9	Dodsworth				

Module 4

Introducing DAX

Section 4.A

Defining DAX

4.A.1 Data Analysis eXpression Language

- DAX stands for Data Analysis eXpression Language
- DAX was developed by Microsoft as the expression language native to the in-memory columnar database used for information analysis
- DAX is used to add meaning to the data that you are analyzing through formulas, or expressions
- DAX is used everywhere xVelocity is used
- Learning how to write expressions with DAX is essential to getting the most out of Power BI

4.A.2 DAX and the Data Model

- How the data that DAX works with is shaped will drastically affect how useful our analytics can be
- DAX is used both to help model data, and to analyze the modeled data
- DAX helps you create new information from data already in your data model
 - ◇ For example
 - * Percentages
 - * Summaries
 - * Key Performance Indicators
 - * Trends
- DAX expressions become a part of your model
- DAX helps you to manage the granularity of your data
 - ◇ It might turn an order level discount into a line item discount for reports that work on the granularity of an order detail row
 - ◇ It helps you define calculations that will summarize data with totals or percentages after that data has been filtered to a subset of itself

4.A.3 Analysis with DAX Calculations

- DAX is used to analyze data
- It is most useful for two types of analysis
 - ◇ Aggregation
 - ◇ Filtering
- When you are dealing with millions of detail records, you need a way to summarize their meaning at various levels or granularities
 - ◇ You want your eyes to be able to see the data in graphical reports without having to add your own meaning to each number
- There are many built in functions that help with various forms of analysis
 - ◇ Statistical Analysis
 - ◇ Time-based Analysis
 - ◇ Dimension slicing and dicing through user defined hierarchies

4.A.4 Calculations not suitable for storage

- DAX can be used to define calculations that are not suitable for storage in the data set
 - ◇ These are values where if data in another row changes, the result of the calculation changes
 - ◇ These are values where the calculation may be performed in many different contexts, where the available contexts are driven by the data itself
- Examples include:
 - ◇ For a given line item of an order, what percentage of that order does the single line item represent?
 - ◇ What was the average age of the customer at the time a sale was made?
 - ◇ Correctly calculated percentages at different grouping levels
 - ◇ Aggregate summaries
- DAX allows us to reshape our data and add expressions so that reporting against our data becomes as easy as checking off boxes

4.A.5 DAX is an Expression Language

- DAX is an **expression language**
 - ◇ A collection of functions, operators, and constants that can be used in a formula or expression to calculate and return one or more values
- Writing DAX feels a little like writing spreadsheet expressions
 - ◇ Excel might add a column by using other values on the same row by referencing data coordinates
 - ◇ Excel might add a summary by using values referenced by a range of data coordinates
 - ◇ DAX does not use co-ordinates to get to its data like Excel; instead DAX expressions reference named identifiers in the data model, like the name of a table or the name of a column
- DAX understands how data is related because of its data model
 - ◇ This makes DAX a much more effective way to analyze data than using Excel expressions

4.A.6 DAX versus Power Query: Similarities

- Both Power Query and DAX have some similarities
 - ◇ Both can be used to calculate column values
 - ◇ Both can be used to calculate entire tables
 - ◇ Both can be used to change a column's name
 - ◇ Both can be used to change a table's name
 - ◇ Either can feel more intuitive than the other for certain situations

4.A.7 DAX versus Power Query: Power Query

- Used to bring data into the model
- Made up of multiple “steps” that each alter the shape of the result, one step at a time.
- Power Query may improve model performance because of query folding
 - ◇ Query folding pushes work into the underlying data source, such as Microsoft SQL Server or cloud database service
 - ◇ This could mean that less data gets transferred into Power BI
 - ◇ This could allow a source DBMS to utilize indexes to perform expensive operations like filtering for data that meets certain criteria, or putting data in a specific order
- Power Query cannot be used to calculate dynamic “measures” outside the context of data in a column
 - ◇ DAX allows creation of measures that are calculated within a filter context, which is not known until the user incorporates the measure into a report

4.A.8 DAX versus Power Query: DAX

- Limited to working with data that has already been brought into the model
- Does not have “steps” but rather is a single series of nested operations
- Feels more like writing spreadsheet expressions
 - ◇ Except DAX uses table and column names instead of cell coordinates
- Can be used to calculate dynamic measures
 - ◇ This allows advanced summary level analysis

4.A.9 DAX versus Power Query: Which?

- Use Power Query to avoid bringing unnecessary data into the model
 - ◇ Dropping unnecessary data at the Power Query phase will help the model perform better and keep the memory footprint smaller
- Use Power Query if you want to re-use queries in importing or transforming data
 - ◇ Query patterns can be cut-and-pasted with modifications as needed to reuse within or across projects
- Only DAX can be used to calculate measures
 - ◇ These powerful calculations need to be recalculated for different filter contexts and values are not stored with the row
- Use DAX when building more complex calculations that require data spanning across tables
 - ◇ DAX can also create role-playing tables that slice or join existing model data, to help tailor the model to analysis needs without duplication

4.A.10 DAX Family of Products

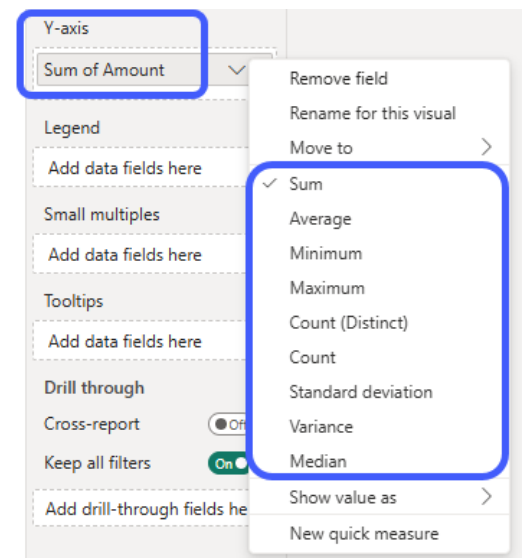
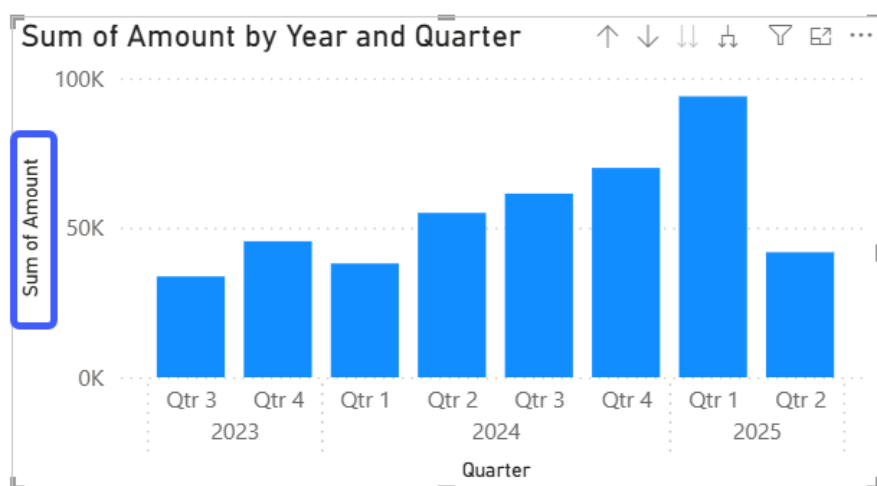
- Power Pivot for Microsoft Excel
 - ◇ Became available as an add-on in 2010
 - ◇ Distributed as a built-in feature starting with Office 2016
- Microsoft Power BI
 - ◇ Locally in Power BI Desktop
 - ◇ In the cloud with Power BI Service
 - ◇ On-premise server with Power BI Report Server
 - ◇ Supports personal, team, and organizational BI Needs
- Microsoft SQL Server Analysis Services - Tabular Mode
 - ◇ Tabular mode is the newer of the two modes supported by SSAS
 - ◇ Became available starting in 2012

Section 4.B

Getting Started with DAX

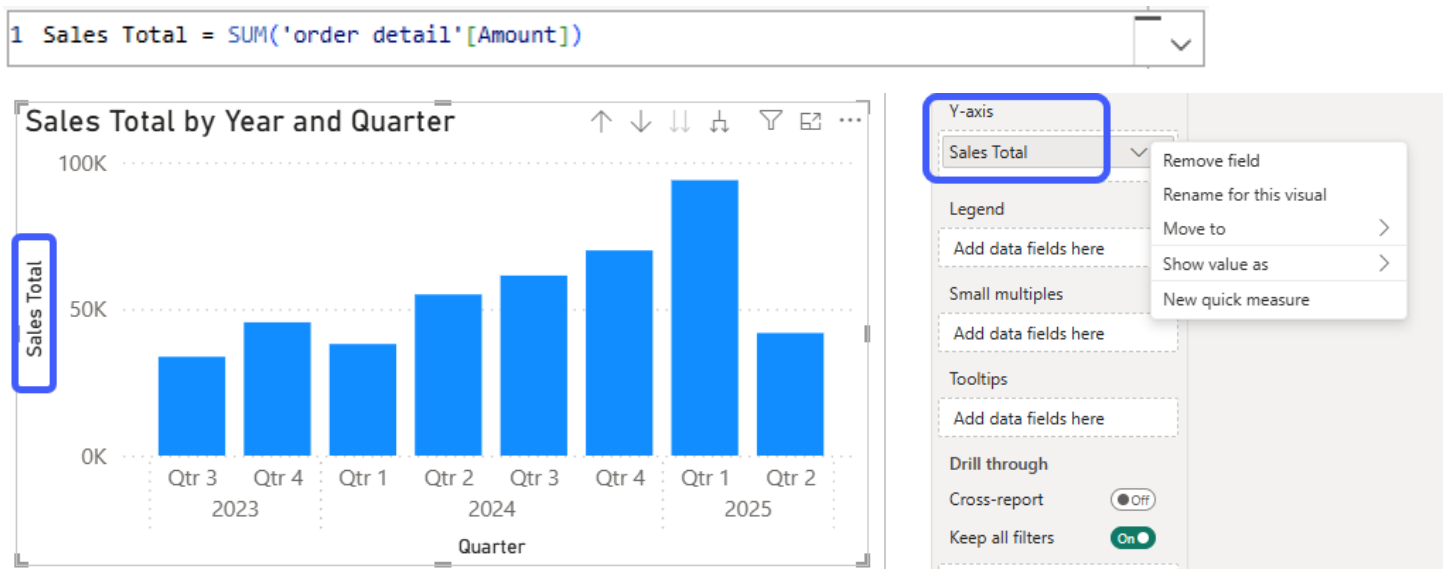
4.B.1 Power BI Creates Implicit DAX Calculations

- When you use a numeric field in a visualization's data well (such as the Y-axis of a column chart), Power BI automatically creates an **implicit measure**, a DAX calculation applied behind the scenes
- The default aggregation is Sum, but this can be modified to other aggregation types such as Average, Count, Min, Max, etc.
- The below example adds up sales using the Amount column from a table named 'order detail', producing a default y-axis label of "Sum of Amount"
 - ◇ This uses the implicit measure `SUM('order detail'[Amount])` (using the Amount column on the 'order detail' table)
 - ◇ We don't see the actual formula `SUM('order detail'[Amount])` anywhere, and the implicit measure can only be modified by clicking to select a different aggregation option:



4.B.2 Creating Explicit DAX Calculations

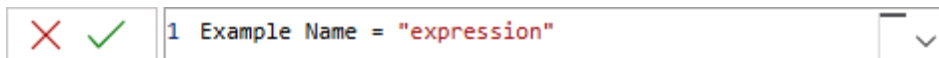
- You can create your own explicit named measures and other calculations to enhance your data model
- Measures are especially helpful:
 - ◇ Where you want to make an explicit calculation and assign it a name
 - ◇ Where you want to be able to use a calculation within another calculation
- A measure can be very simple or very complex
 - ◇ Here is an update to the previous column chart that instead uses the explicit measure [Sales Total] , defined using the calculation `SUM('order detail'[Amount])`



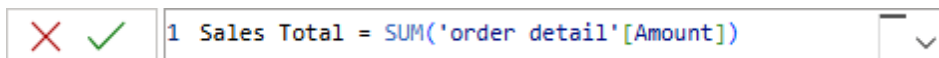
- ◇ Because this is an explicit measure, there is no click-to-select option to change the aggregation type used; instead, you would directly modify the DAX calculation used in the measure

4.B.3 Writing DAX Expressions

- DAX calculations are written in the **formula bar** in Power BI, which appears on choosing to create a new measure, new column, or new table
- DAX expressions can include hard-coded values, mathematical expressions, or use of functions from the DAX library
- Adding a new measure, column, or table with DAX starts by providing a name for the new object followed by an equal sign, then the DAX formula



- Within the DAX formula, any referenced field names (columns or measures) must appear in square brackets, and a fully-qualified name starts with the table that the field belongs to



- ◇ The table name must be in single quotes if it contains a space, but field names do not use quotes (only brackets)
- Clicking the green check in the formula bar (or hitting enter) will save the DAX expression to the model

4.B.4 What is a Measure?

- A **measure** is a calculation that is saved to the model and is only run when it is called
 - ◇ Measures can be used directly in report visualizations
 - ◇ A measure can also be used by another measure, sort of like a variable

- **Examples:**

- ◇ These are all measures based on a table named 'order detail':

```
Sales Total = SUM('order detail'[Amount])
```

```
# of Orders = COUNTROWS('order detail')
```

```
Avg Order Value = [Sales Total] / [# of Orders]
```

- ◇ A measure can also be used to save a hard-coded value ...

```
Annual Sales Goal = 1000000
```

```
Report Last Updated = "March 1, 2026"
```

- ◇ ... or other calculation

```
Monthly Sales Goal = [Annual Sales Goal] / 12
```

4.B.5 What is a Calculated Column?

- A **calculated column** uses DAX to add a new column on a table in the data model

- ◇ Unlike a measure which is only run at the time it is used somewhere, a calculated column runs at the time it is created (or refreshed)
- ◇ The calculated row-level values are then saved to the data model, and the resulting column can be used just like any other column

- **Examples**

- ◇ The below expression could be used to create a new column on the 'order detail' table named [Sales Tax (7%)], where the row value is produced by taking the corresponding value in the [Amount] column and multiplying by 7%:

```
Sales Tax (7%) = 'order detail'[Amount] * .07
```

- ◇ The above expression would *not* work to create a measure because there is no summarization function to return a single value for [Amount]; instead, this calculation cycles through each row-level [Amount] value for each record in the table

4.B.6 Common Aggregation Functions in DAX

- A few commonly used functions for summarizing data include:
 - ◇ SUM()
 - ◇ COUNT(), COUNTA(), COUNTROWS()
 - ◇ AVERAGE()
 - ◇ MIN(), MAX()
- DAX also includes many, many other types of functions including functions to filter data (FILTER), change filter evaluation context (CALCULATE, ALL), reach across to related tables (RELATED, USERELATIONSHIP), and perform periodized date calculations, which will be explored further in Week 10

Module 5

Visualizing Data

Section 5.A

Working with Visualizations

5.A.1 Choosing Visuals

- The shape of the visualization should support the story
 - ◇ Consider the level of detail needed on the page
 - ◇ Also consider what interactions you want to make available to the report's end user
- Prioritize clarity of presentation and interaction
 - ◇ How will your audience be accessing the report information?
 - ◇ What context will they need to understand and interact with the visuals?
- Accessibility (font size, image size, use of color)
 - ◇ Assume that your users will be viewing the report on a screen smaller than the one you are using to create it.
 - ◇ Use accessible color schemes (built in under View tab) or check custom color schemes for visual clarity
 - * Tools like DaltonLens desktop software or Adobe's "Color Blind Safe" online accessibility checker can help you ensure that your color palette is unambiguous (<https://daltonlens.org/>, <https://color.adobe.com/create/color-accessibility>)
 - ◇ Consider using **labels**, **icons**, and different line styles (dotted, dashed) to augment the meaning of colors in your design
 - * Labels may deliver a better experience with screen-reader apps
 - * Icons and textures may support better printing output, especially if printed in grayscale

5.A.2 Configuring Visuals

- Available fields vary by type of visualization
 - ◇ X-axis is often categorical/qualitative
 - ◇ Y-axis or Values is often numeric/quantitative (or uses count)
 - ◇ Legend is typically categorical/qualitative
 - ◇ Small multiples creates multiple parallel versions of a chart
 - ◇ Tooltip adds info available on hover
- Visuals are configured by default with some basic format settings, but you may want to adjust these
 - ◇ Are the color choices unambiguous?
 - ◇ Are the meanings of labels clear? Are they readable?
 - ◇ Is the format of the visual useful?
- Consider the order and arrangement of visuals
- You can also add non-data visuals such as headers, images, and navigation buttons to enhance your report

5.A.3 Visuals Need Good Data

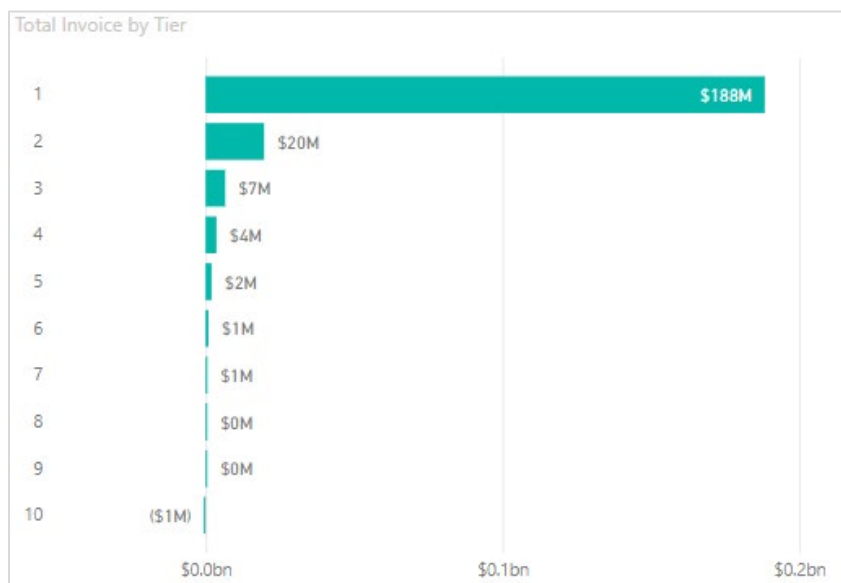
- Consider whether you need to revisit how the data is shaped to create the visualizations you want
 - ◇ Additional measures and calculated columns may be needed to support your visuals.
 - ◇ Adjustments to the data model may be needed to add or manage relationships.
- For large data sets, there may be performance considerations based on type and number of visuals
- It's very common to discover issues with your data once you start creating visualizations, so just consider that part of the process!

5.A.4 Types of Visualizations

- Visuals to aggregate and compare data
 - ◇ bar and column charts
 - ◇ donut and pie charts
 - ◇ line and area charts
 - ◇ scatter charts
 - ◇ maps
- Highlight a specific data point or progress against a goal
 - ◇ cards
 - ◇ gauge charts
 - ◇ KPIs
- Other non-data visuals

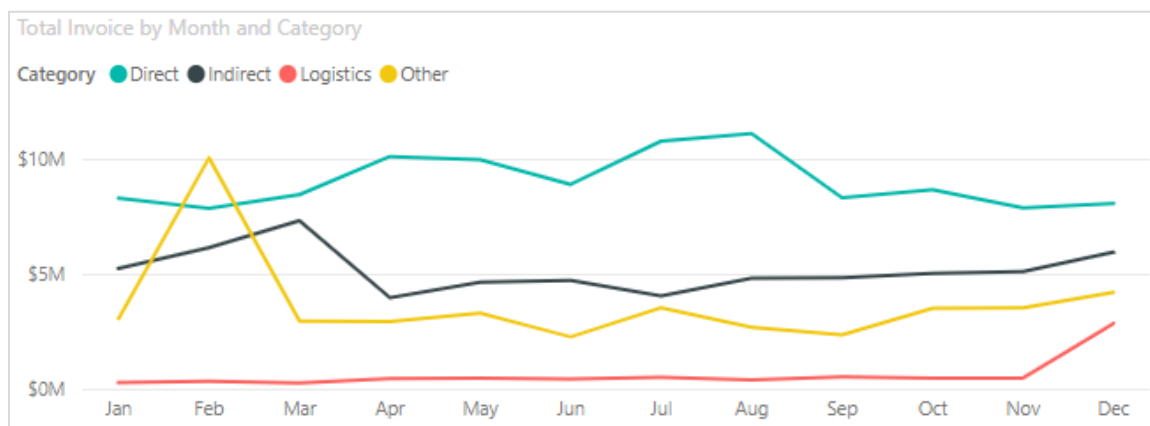
5.A.5 Bar charts and column charts

- These are simple and powerful visuals for comparing values across categories.
- ◇ Sub-values in a category can be stacked or clustered

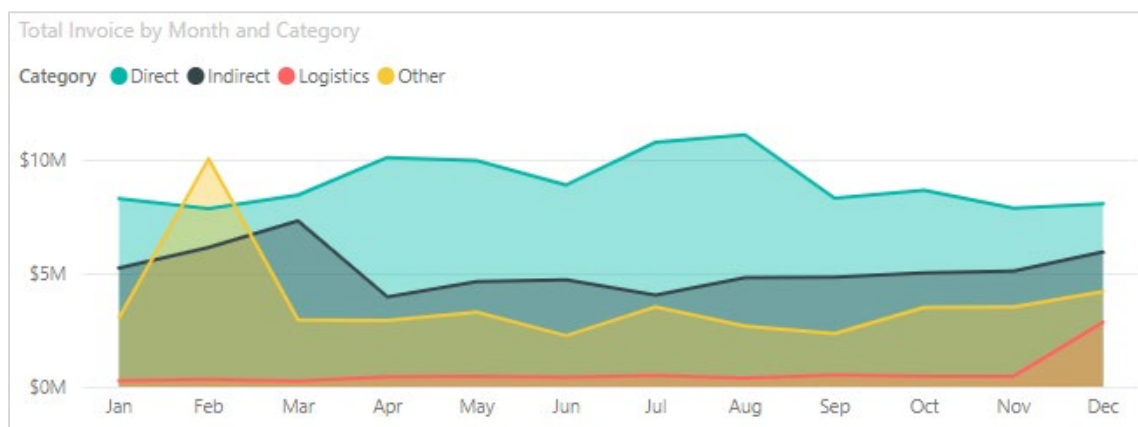


5.A.6 Line charts and area charts

- Line charts trace the shape of a series of values over time to show trends

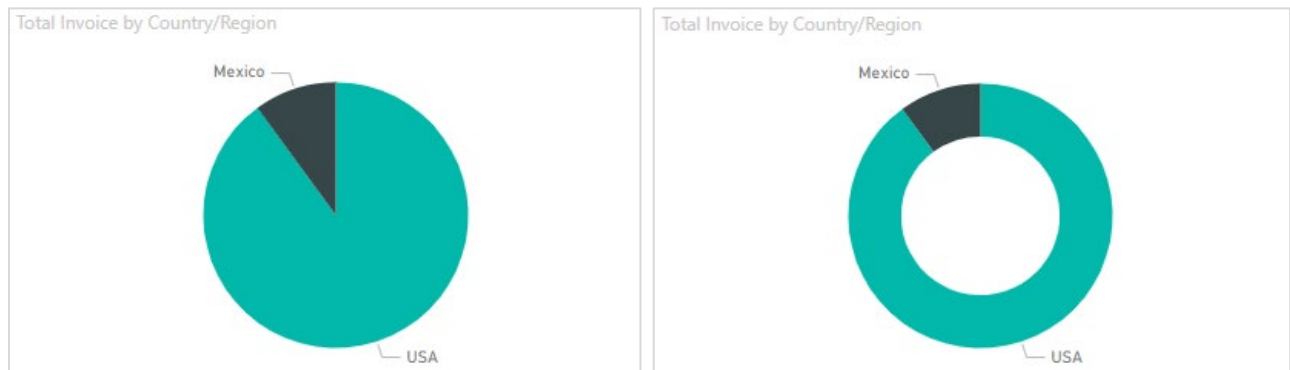


- Area charts fill in that shape to emphasize the magnitude of change over time.

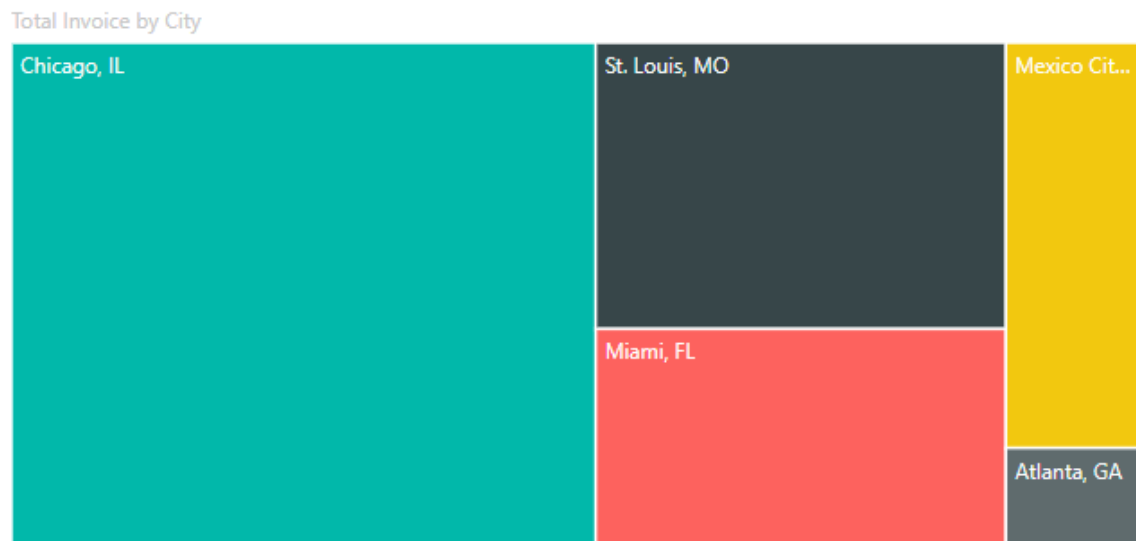


5.A.7 Pie charts, donut charts, and treemaps

- These charts compare parts within a whole
 - ◇ A doughnut chart is just like a pie chart, but with a blank hole in the middle



- ◇ A treemap displays relative values as rectangles within a rectangle, with the largest values starting at the top left and arranged down to the smallest at the bottom right



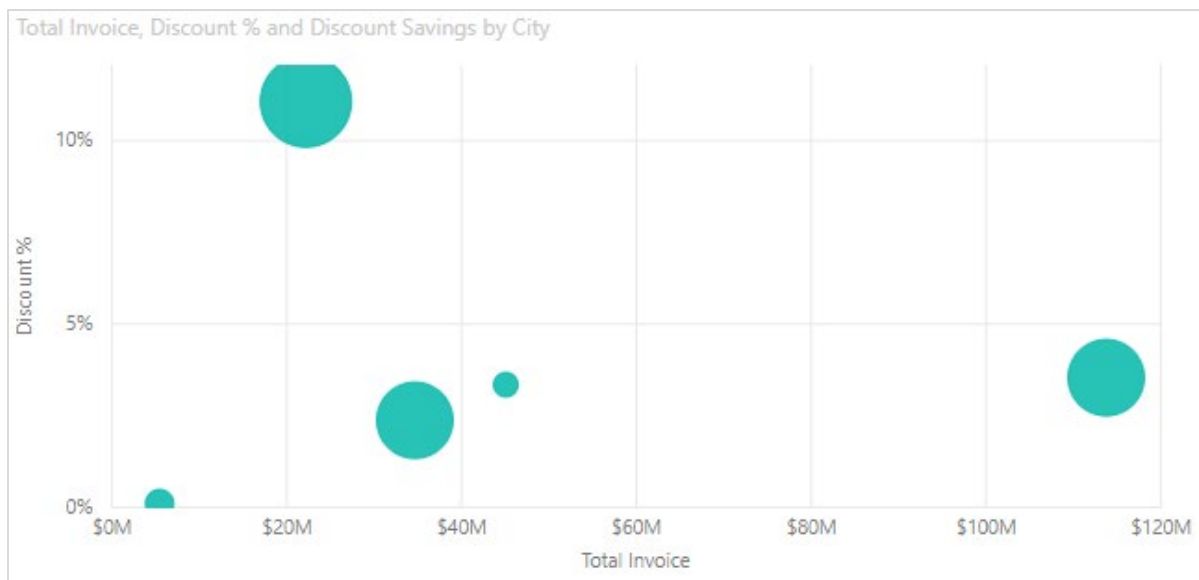
- ◇ The “whole” should add up to 100% (no overlap)

5.A.8 Scatter and bubble charts

- A scatter chart plots data points against an X and Y axis that are both numerical values, and is useful for showing clusters or trends in data.



- A bubble chart sizes the data points based on an additional dimension of value.



5.A.9 Table and Matrix

- As a visual, tables are a way to quickly see and compare detailed data by category.

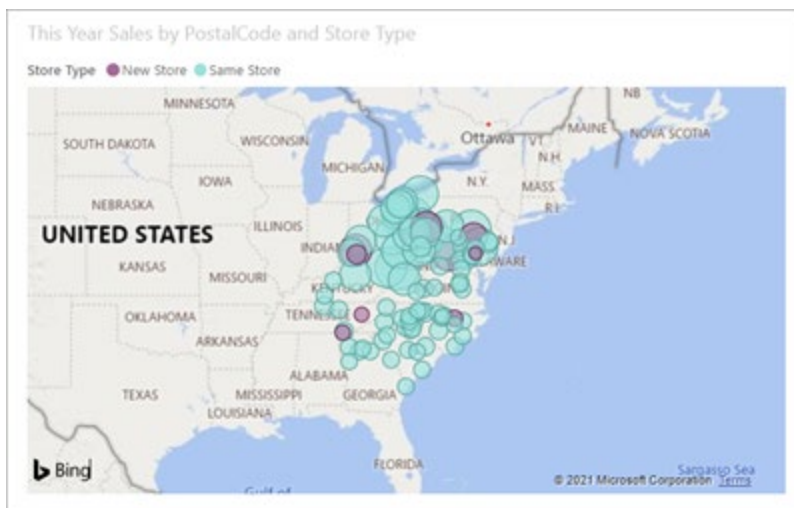
Category	Sub Category	Total Invoice
Direct	Contracting & Services	\$3,556,021.52
Direct	Hardware	\$140,732,213.09
Direct	Outsourced	\$29,429,955.04
Direct	Raw Materials	\$42,965,010.75
Indirect	Indirect Goods & Services	\$156,848,846.62
Logistics	Logistics	\$23,939,576.96
Other	Other	\$119,863,903.91
Total		\$517,335,527.89

- A matrix shows the breakdown and subtotals of data across dimensions.

Category	Contracting & Services	Hardware	Indirect Goods & Services	Logistics	Other	Outsourced	Raw Materials	Total
Direct	\$3,556,021.52	\$140,732,213.09				\$29,429,955.04	\$42,965,010.75	\$216,683,200.40
Indirect			\$156,848,846.62					\$156,848,846.62
Logistics				\$23,939,576.96				\$23,939,576.96
Other					\$119,863,903.91			\$119,863,903.91
Total	\$3,556,021.52	\$140,732,213.09	\$156,848,846.62	\$23,939,576.96	\$119,863,903.91	\$29,429,955.04	\$42,965,010.75	\$517,335,527.89

5.A.10 Basic Maps

- Map visuals provide a quick way to associate both categorical and quantitative information with a location.
- A basic map can be used to plot individual data points or bubbles sized on a value dimension.



5.A.11 Cards

- Card visualizations can be used to display a single data point or multiple specific data points

439.10K

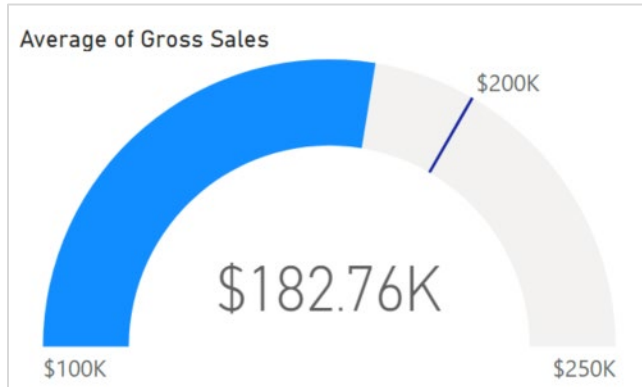
Sales Total

324.54

Avg Order Value

5.A.12 Scorecards & Metrics

- A gauge chart typically displays progress toward a goal as a shaded arc, with the target value or goal represented by a single line



- The KPI visualization displays progress made toward a key performance indicator defined by the business



- These are typically used:
 - ◇ To check progress on key measures
 - * *What are we ahead / behind on?*
 - ◇ To assess distance to a target metric
 - * *How far is this ahead / behind?*

5.A.13 Non-Data Visual Elements

- Images, text boxes, forward and back buttons, links, and other visual elements can also be added to a report from the **Insert** tab in the report view
 - ◇ Create user-friendly labeling and navigation
 - ◇ Incorporate brand identity
 - ◇ Add other non-data visual elements

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5.A.14 Week 9 Wrap-Up

- This week introduced Power BI as the final major tool in your data analyst toolkit
 - ◇ Module 1 established the Power BI ecosystem and the xVelocity engine at its core. You toured the interface and learned how Power BI Desktop, Power Query, and DAX all fit together
 - ◇ Module 2 introduced the semantic model and the principles behind a well-designed one: data (fact) tables and lookup (dimension) tables; star and snowflake schema rules; and what goes into modeling
 - ◇ Module 3 covered Power Query as the transformation layer: the tool that brings raw data into your model, shapes it, and cleans it before modeling begins
 - ◇ Module 4 introduced DAX, the expression language that powers calculations in the model; the basics of implicit and explicit measures; and the key distinction between measures (evaluated on demand within a filter context) and calculated columns (stored in the model row by row)
 - ◇ Module 5 connected the model to the report layer, covering how to choose an appropriate chart type, configure visuals, and consider accessibility and audience — turning your data into a story that others can read and interact with
- Coming up in Week 10:
 - ◇ On the Power Query side, you will move beyond the GUI and begin writing M language expressions directly
 - ◇ On the DAX side, you will explore filter context in detail, including the CALCULATE function, time intelligence functions for period-over-period analysis, and techniques for writing measures that answer more complex analytical questions

End of Week 9 Student Guide